

Music and Math

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Chapter 1

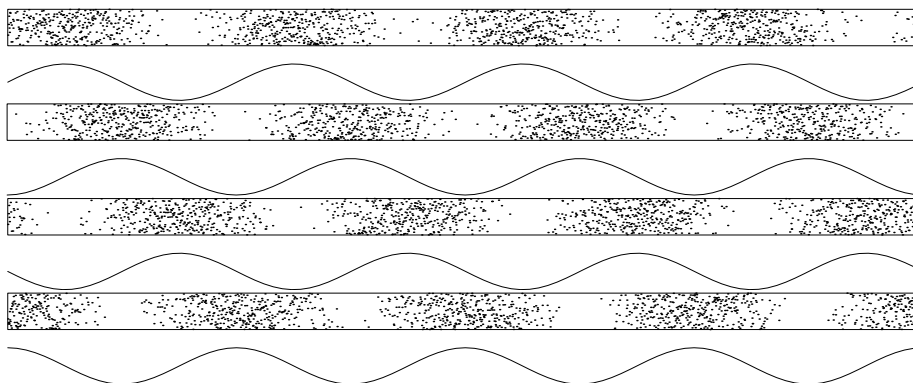
Sound

Definition 1 (Sound). *Sound is a mechanical wave that propagates through a medium, typically air or water.*

Definition 2 (Wave). *A wave is an oscillation in a medium.*

Definition 3 (Longitudinal waves). *Longitudinal waves are waves where the displacement of the medium is parallel to the the direction the wave travels.*

Definition 4 (Transverse waves). *Transverse waves are waves where the displacement of the medium is perpendicular to the the direction the wave travels.*



1.1 Homework

1. Are sound waves transverse or longitudinal waves?
2. When representing waves graphically do draw them as transverse or longitudinal waves?

Chapter 2

Trigonometry

We will only need the use of two of the six standard trigonometric functions: $\sin \theta$ and $\cos \theta$. These functions are defined in terms of ratios of sides of a right triangle. Note that unless told otherwise all of our triangles will be right triangles. So let us first discuss some facts about triangles and right triangles in particular. Let us start with the definition of a triangle.

Definition 5 (Triangle). *A triangle is a polygon with three sides and three angles.*

Next let us define a right triangle.

Definition 6 (Right triangle). *A right triangle is a triangle with a right, 90° , angle. The side opposite the right angle is called the hypotenuse and the other two sides are called legs.*

The following result is true for any triangle.

Theorem 1 (Sum of angles in a triangle). *The sum of the angles of a triangle equals 180° .*

If we have a right triangle then the following result holds.

Theorem 2. *In a right triangle the sum of the two non-right angles equals 90° .*

All right triangles satisfies the Pythagorean theorem:

Theorem 3 (Pythagorean theorem). *The sum of the squares of the legs of a right triangle is equal to the square of the hypotenuse.*

Using the labels in figure 2.1 we have $A^2 + O^2 = H^2$. So if we are given any two sides of a right triangle we can solve for the third.

Example 1. ???EXAMPLE Pythagorean theorem GOES HERE???

If the angle θ is placed in the lower left position as shown in figure 2.1 then the leg of the triangle that forms the lower half of angle θ is called the adjacent side and we will label it as A . The other leg is referred to as the opposite side and is labeled as O . The side opposite the right angle is called the hypotenuse and is labeled H .

The sine function is defined as the ratio opposite side to hypotenuse.

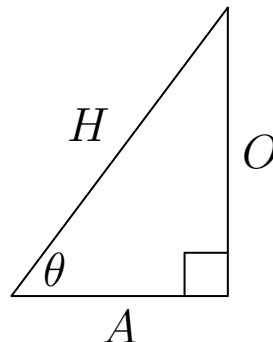


Figure 2.1: Right triangle with adjacent, opposite, and hypotenuse

Definition 7 (Sine). *If in a right triangle the leg opposite to angle θ is O and the hypotenuse is H , then sine of θ is defined by*

$$\sin \theta = \frac{O}{H}.$$

The cosine function is defined to be the ratio adjacent to hypotenuse,

Definition 8 (Cosine). *If in a right triangle the leg adjacent to angle θ is A and the hypotenuse is H , then cosine of θ is defined by*

$$\cos \theta = \frac{A}{H}.$$

2.1 Sine for acute angles

There are a few angles that are very convenient to use when doing trigonometry: 30° , 45° , and 60° . The triangles associated with them are the 45° - 45° - 90° triangle and the 30° - 60° - 90° triangle seen in figure 2.2. Using these two triangles we will be able to calculate $\sin \theta$ for the angles:

30° , 45° , 60° , 120° , 135° , 150° , 210° , 225° , 240° , 300° , 315° , and 330° .

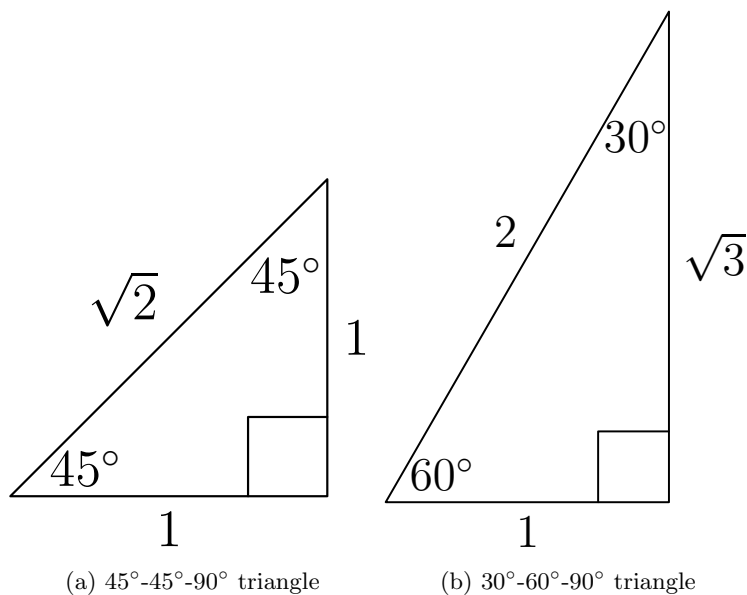


Figure 2.2: Special angle triangles

To evaluate $\sin 45^\circ$ we look at figure 2.2a and note that the length of the side opposite the 45° angle is 1 and the hypotenuse is $\sqrt{2}$. Since by definition

$$\sin \theta = \frac{O}{H} \text{ we have } \sin 45^\circ = \frac{1}{\sqrt{2}}.$$

Suppose we want to find $\sin 60^\circ$. First we note that in figure 2.2b the side opposite the 60° angle is $\sqrt{3}$ and that the hypotenuse is 2. Again by definition

$$\sin \theta = \frac{O}{H} \text{ so we have } \sin 60^\circ = \frac{\sqrt{3}}{2}.$$

To calculate $\sin 30^\circ$ we again look at figure 2.2b, but this time we see that the side opposite the 30° angle is 1 while the hypotenuse is still 2 so

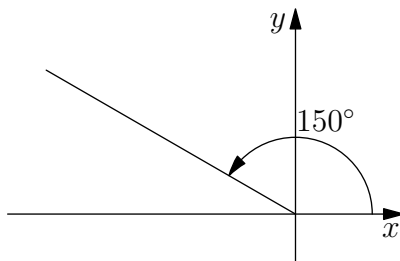
$$\sin 30^\circ = \frac{1}{2}.$$

2.2 Sine for non-acute angles

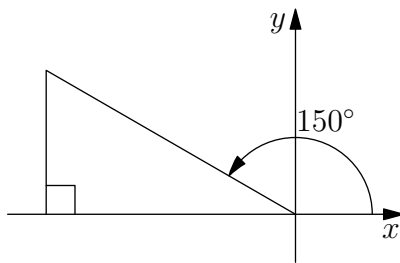
Let us remind ourselves of what standard position of an angle is.

Definition 9 (Standard position). *An angle θ is in standard position if its vertex is at the origin of the coordinate system and its initial side is on the positive x -axis.*

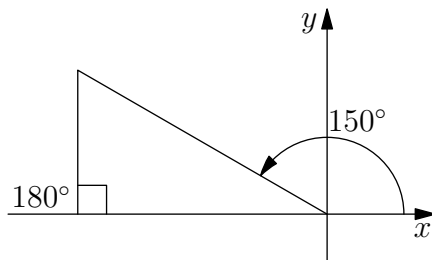
Suppose we wish to find $\sin 150^\circ$. The angle 150° is no longer in the first quadrant so it will take an extra step to calculate the sine of it. We first draw the angle in standard position:



Next we drop a perpendicular from the angle down to the x -axis.



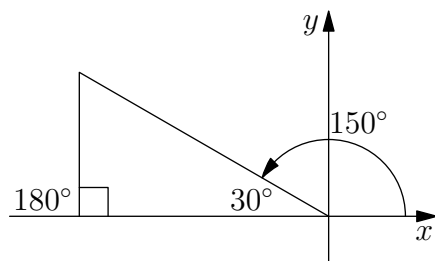
This will always result in the construction a triangle. We find the angle of this new triangle located at the origin and we need to determine its size. There are at least two ways to do this, but they both start the same way. First we note that horizontal leg is located at 180° .



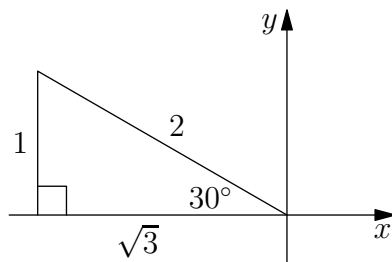
Now we need to figure out how far it is from 150° to 180° . We can do this by either

1. Subtracting $180^\circ - 150^\circ = 30^\circ$
2. or asking how much farther past 150° I must go to get to 180° , its 30° .

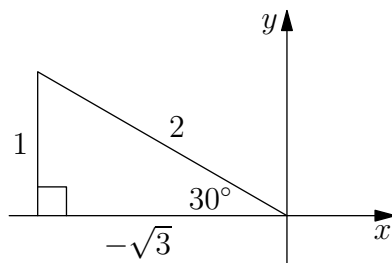
Either way we find the the angle we were looking for is 30° , called the reference angle:



This means we have a 30° - 60° - 90° triangle and so it has side lengths shown below:



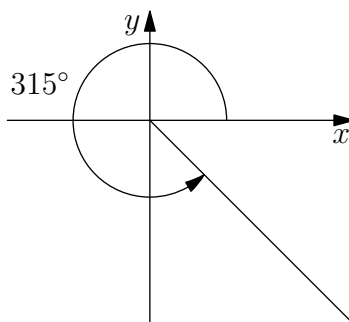
The last step in determining $\sin 150^\circ$ is to put the correct signs on the sides of our triangle. Since we are on the left of the y -axis the x values are negative and since we are above the x -axis the y values remain positive. The hypotenuse is always *positive*. This gives us our final drawing:



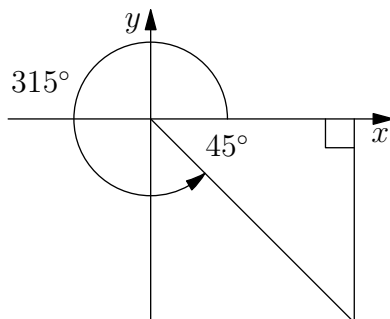
Using the reference angle, the 30° we found, we can determine $\sin 150^\circ$. Since the side opposite the 30° angle is 1 and the hypotenuse is 2 and by definition

$$\sin \theta = \frac{O}{H} \text{ we have } \sin 150^\circ = \frac{1}{2}.$$

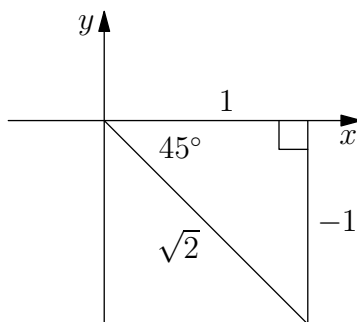
Let us calculate $\sin 315^\circ$. First we draw the angle 315° in standard position:



Next we draw the triangle for this angle and find the reference angle:



Finally we fill in the sides of this triangle with the appropriate signs:



Since sine is opposite over hypotenuse we find that:

$$\sin 315^\circ = \frac{-1}{\sqrt{2}}$$

2.3 Sine for quadrantal angles

If an angle is a quadrantal angle (0° , 90° , 180° , and 270°), then there is no triangle that can be drawn. Instead we will use the unit circle shown in figure 2.3 below.

θ	0°	90°	180°	270°	360°
point	$(1, 0)$	$(0, 1)$	$(-1, 0)$	$(0, -1)$	$(1, 0)$
$\sin \theta$	0	1	0	-1	0

Table 2.1: Sine table of quadrantal angles.

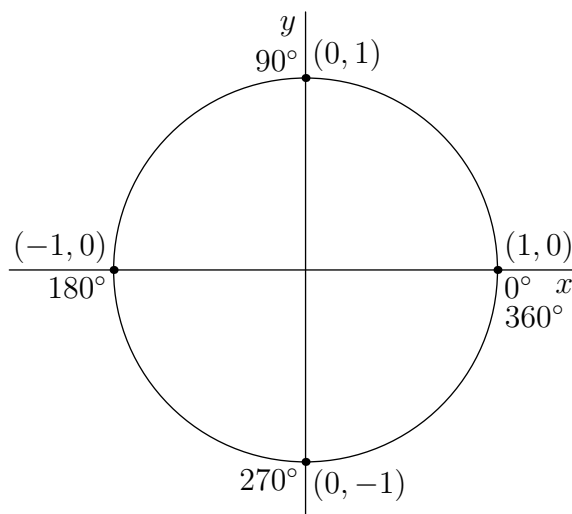


Figure 2.3: Circle with quadrantal angles.

When using the unit circle the $\sin \theta$ is the y -coordinate of the point corresponding to that angle. So to find $\sin 0^\circ$ we look at the unit circle and see that the point that corresponds to 0° is $(1, 0)$. Recalling that points are always written with the x -coordinate first and y -coordinate second, so (x, y) , we see that the y -coordinate of $(1, 0)$ is 0. Therefore we have

$$\sin 0^\circ = 0.$$

Going around the circle we find the values of $\sin \theta$ for the quadrantal angles are:

2.4 The graph of sine

Instead of thinking about angles rotating about a central point and measuring this in degrees, which is the natural thing to do, we can instead mark off a line into equal pieces and let these represent the angles also measured in degrees.

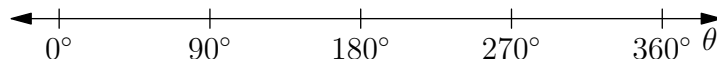


Figure 2.4: Linear representation of angles in degrees.

Now if we plot the values of sine for the quadrantal angles in table 2.1 we get the graph shown in figure 2.5.

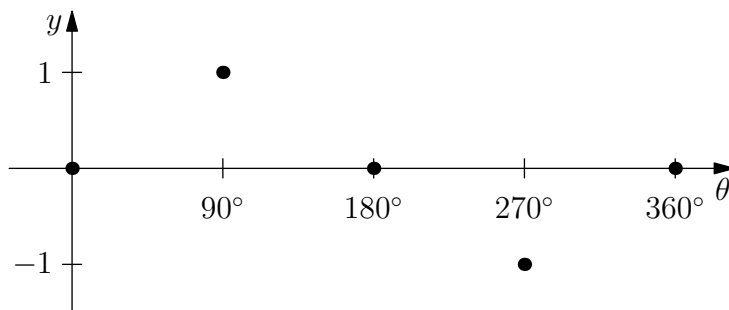


Figure 2.5: Plotting the points of $\sin \theta$ for the quadrantal angles in degrees.

Connecting these points with a smooth curve give us the graph of the sine wave below.

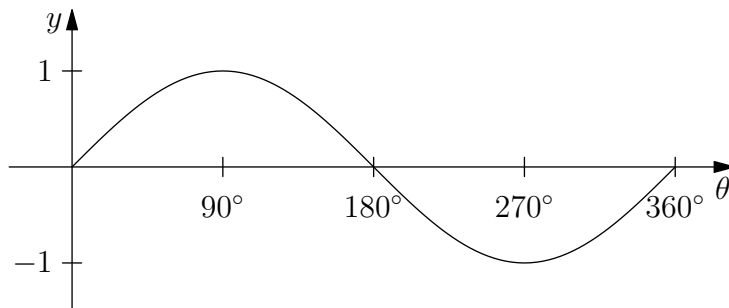
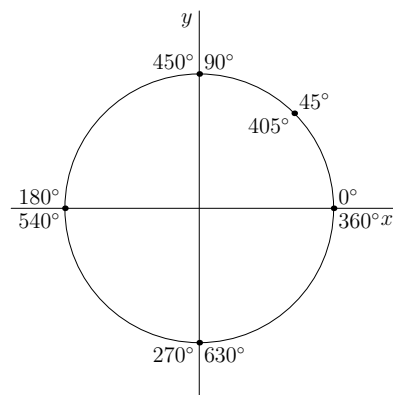


Figure 2.6: Graph of $\sin \theta$ in degrees.

Note that if we were to start at our first quadrantal angle, 0° , and add 90° to it we get our second quadrantal angle 90° . Adding 90° to that gets our third quadrantal angle 180° , another 90° is 270° the fourth quadrantal angle, and finally our last quadrantal angle 360° is one more 90° rotation. What if we add 90° to 360° ? That will put us at 450° . But that just puts us back to the same spot 90° took us to. In fact we have already seen this with 0° and 360° . In terms of our location on the circle we are in the same position if we are at 0° or 360° . The same thing can happen for angles that are not quadrantal angles. For example, 45° and $405^\circ = 360^\circ + 45^\circ$ are located at the same position on the circle.

This property is important enough that we give these angles a name.



Definition 10 (Coterminal angles). *Two angles are coterminal angles if one is an integer multiple of 360° added to the other.*

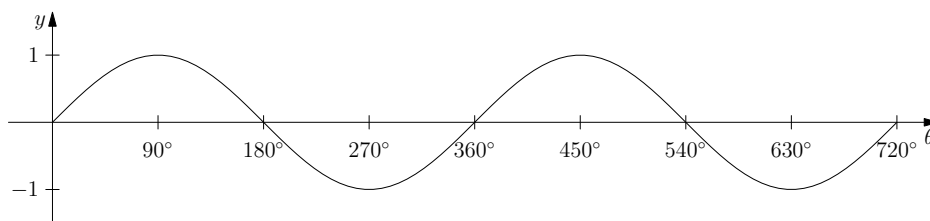
Let find some coterminal angles to 30° .

n	$30^\circ + n \cdot 360^\circ$
0	30°
1	390°
2	750°

And since 30° and 390° are angles that are physically in the same position then they will form the same triangles in the coordinate system. This means that

$$\sin 30^\circ = \sin 390^\circ = \frac{1}{2}.$$

The consequence of this is that the sine wave repeats itself every 360° . Let us graph $\sin \theta$ for θ from 0° to 720° .



Example 2. *Let us calculate the following values:*

1. $\sin 420^\circ$
2. $\sin 510^\circ$
3. $\sin 765^\circ$

1. To find $\sin 420^\circ$ we first note that $420^\circ > 360^\circ$. We want to find the coterminal angle to 420° that is between 0° and 360° . So we subtract 360° from 420° to get $420^\circ - 360^\circ = 60^\circ$. Since 60° is less than 360° and greater than 0° we can find

$$\sin 420^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2}.$$

2. To calculate $\sin 510^\circ$ we do the same thing as the last example, we see that $510^\circ > 360^\circ$ and so to find the coterminal angle we do the subtraction $510^\circ - 360^\circ = 150^\circ$. Since $0^\circ < 150^\circ < 360^\circ$ our answer is

$$\sin 510^\circ = \sin 150^\circ = \frac{1}{2}.$$

3. To find the coterminal angle of 765° will require a an extra step. We first do our subtraction $765^\circ - 360^\circ = 405^\circ$, but this is still greater than 360° so we will subtract 360° again. Now $405^\circ - 360^\circ = 45^\circ$ and that is between 0° and 360° . So

$$\sin 765^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}.$$

2.4.1 Combining the graphs of sine functions

How do we add two sine functions together? Suppose we want to add $\sin \theta$ to itself

$$y = \sin \theta + \sin \theta.$$

Let us look at some representative values of θ and see what happens

θ	$\sin \theta$	$\sin \theta$	$\sin \theta + \sin \theta$
0°	0	0	$0 + 0 = 0$
90°	1	1	$1 + 1 = 2$
180°	0	0	$0 + 0 = 0$
270°	-1	-1	$-1 - 1 = -2$
360°	0	0	$0 + 0 = 0$

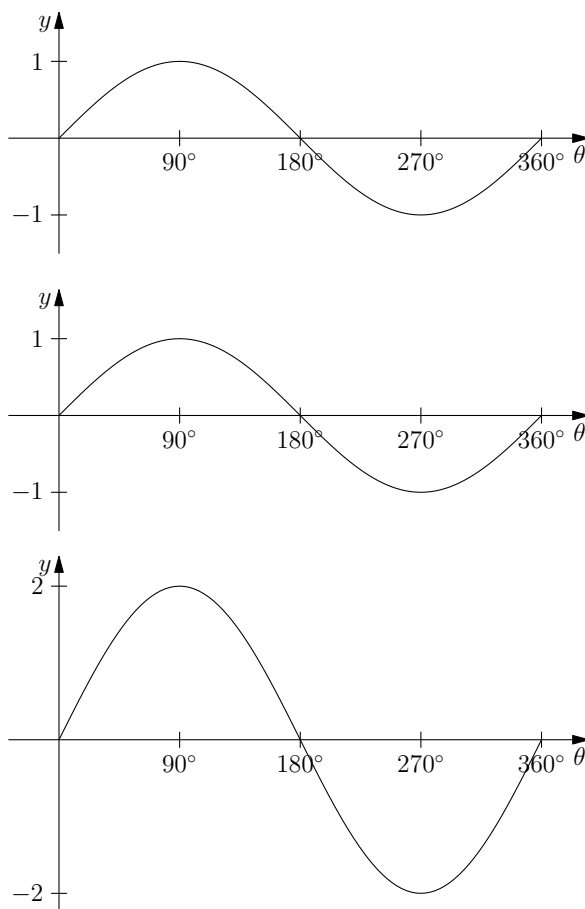


Figure 2.7: Graph of $\sin \theta + \sin \theta$.

This should make sense since $\sin \theta + \sin \theta = 2 \sin \theta$

Next let us graph $y = \sin \theta + \sin 2\theta$.

θ	$\sin \theta$	$\sin 2\theta$	$\sin \theta + \sin 2\theta$	
0°	0	0	$0 + 0 =$	0
45°	.707	1	$.707 + 1 =$	1.707
90°	1	0	$1 + 0 =$	1
135°	.707	-1	$.707 - 1 =$	-0.293
180°	0	0	$0 + 0 =$	0
225°	-.707	1	$-.707 + 1 =$	0.293
270°	-1	0	$-1 + 0 =$	-1
315°	-.707	-1	$-.707 - 1 =$	-1.707
360°	0	0	$0 + 0 =$	0

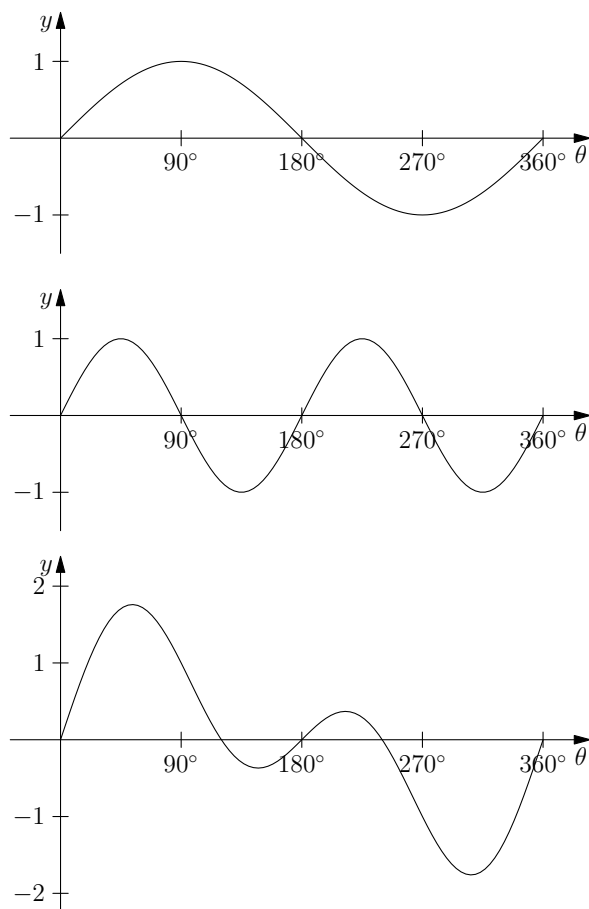


Figure 2.8: Graph of $\sin \theta + \sin 2\theta$.

Now let us graph $y = \sin \theta + .5 \sin 2\theta$.

θ	$\sin \theta$	$.5 \sin 2\theta$	$\sin \theta + \sin 2\theta$	
0°	0	0	$0 + 0 =$	0
45°	.707	.5	$.707 + .5 =$	1.207
90°	1	0	$1 + 0 =$	1
135°	.707	-.5	$.707 - .5 =$	0.207
180°	0	0	$0 + 0 =$	0
225°	-.707	.5	$-.707 + .5 =$	-.207
270°	-1	0	$-1 + 0 =$	-1
315°	-.707	-.5	$-.707 - .5 =$	-1.207
360°	0	0	$0 + 0 =$	0

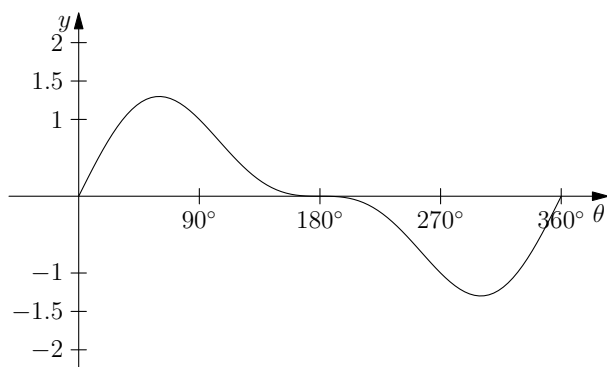
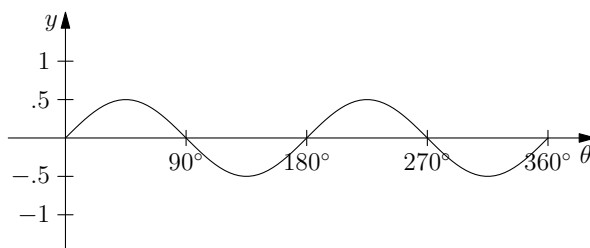
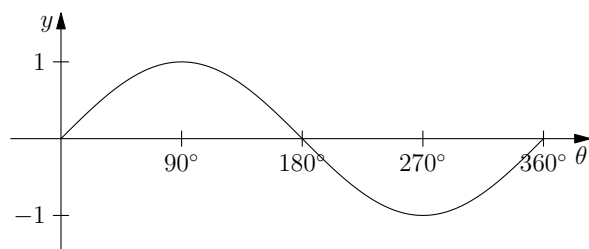
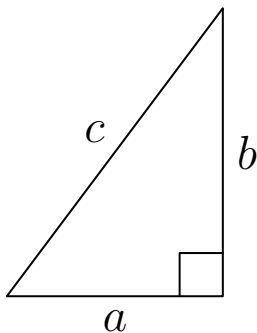


Figure 2.9: Graph of $\sin \theta + .5 \sin 2\theta$.

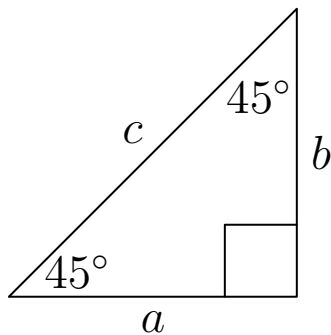
2.5 Homework

1. Given the two sides of the right triangle below find the other side:



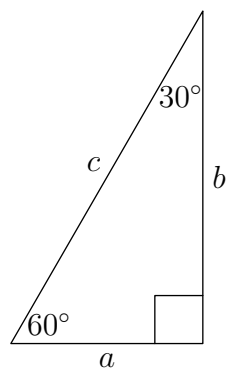
- | | | |
|----------------------|---------------------------|---------------------------|
| (a) $a = 3, b = 4$ | (f) $a = 1, b = 1$ | (k) $b = \sqrt{3}, c = 2$ |
| (b) $a = 5, c = 12$ | (g) $a = 1, c = \sqrt{2}$ | (l) $a = 4, b = 5$ |
| (c) $b = 24, c = 25$ | (h) $b = 1, c = \sqrt{2}$ | (m) $b = 1, c = 10$ |
| (d) $a = 8, c = 17$ | (i) $a = 1, c = 2$ | |
| (e) $b = 20, c = 25$ | (j) $a = 1, b = \sqrt{3}$ | |

2. Given the side below find the other two sides:



- | | | |
|---------------------|-----------------------|-----------------------|
| (a) $a = 3$ | (f) $c = 2\sqrt{2}$ | (i) $c = 1$ |
| (b) $a = 7$ | (g) $a = \frac{1}{2}$ | (j) $c = 3$ |
| (c) $b = 4$ | (h) $b = \frac{3}{2}$ | (k) $c = \frac{1}{2}$ |
| (d) $b = 2$ | | |
| (e) $c = 6\sqrt{2}$ | | |

3. Given the side below find the other two sides:



(a) $a = 2$

(e) $c = 6$

(i) $b = 1$

(b) $a = 5$

(f) $c = 3$

(j) $b = 3$

(c) $b = 4\sqrt{3}$

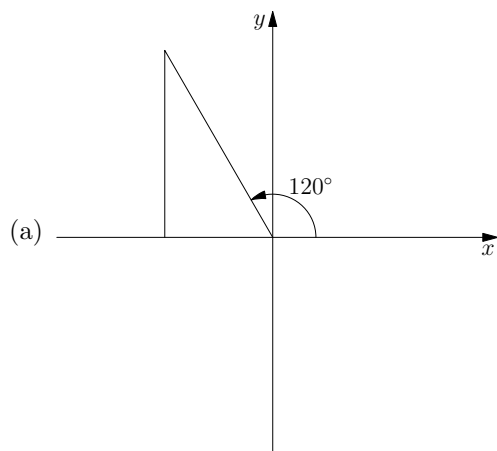
(g) $a = \frac{1}{2}$

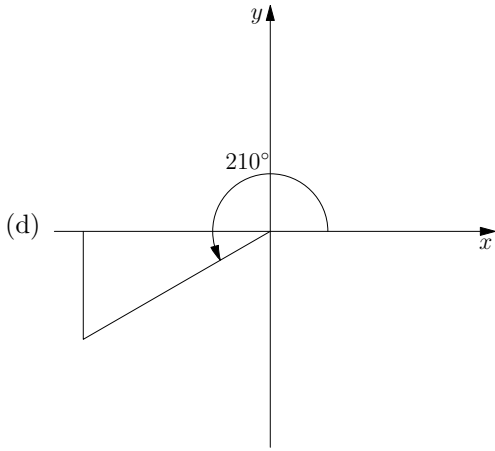
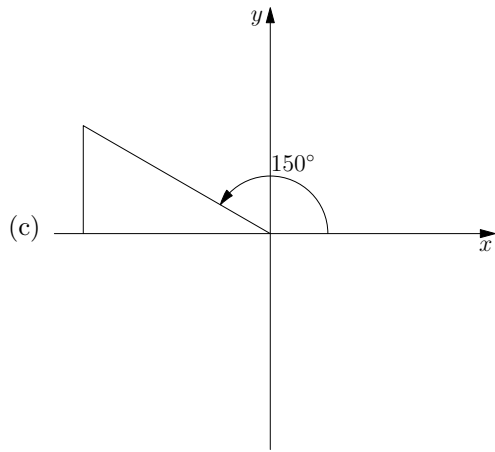
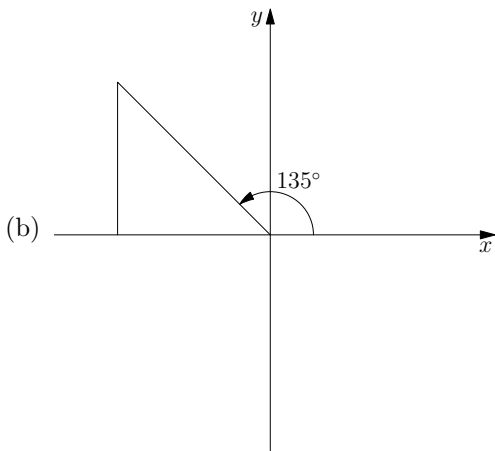
(k) $b = \frac{1}{2}$

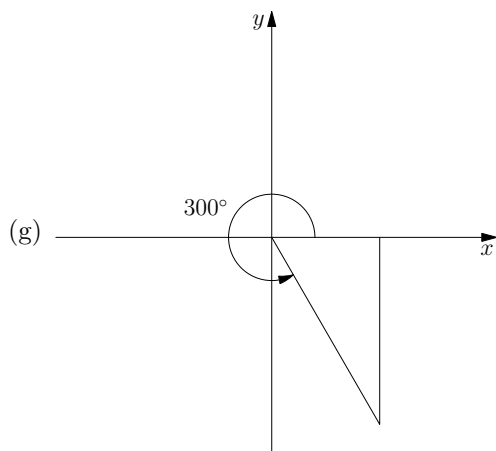
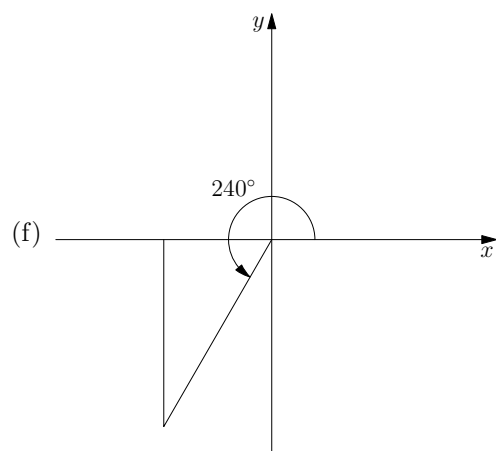
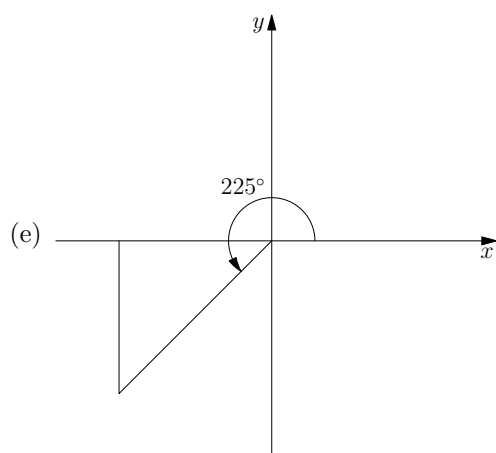
(d) $b = 7\sqrt{3}$

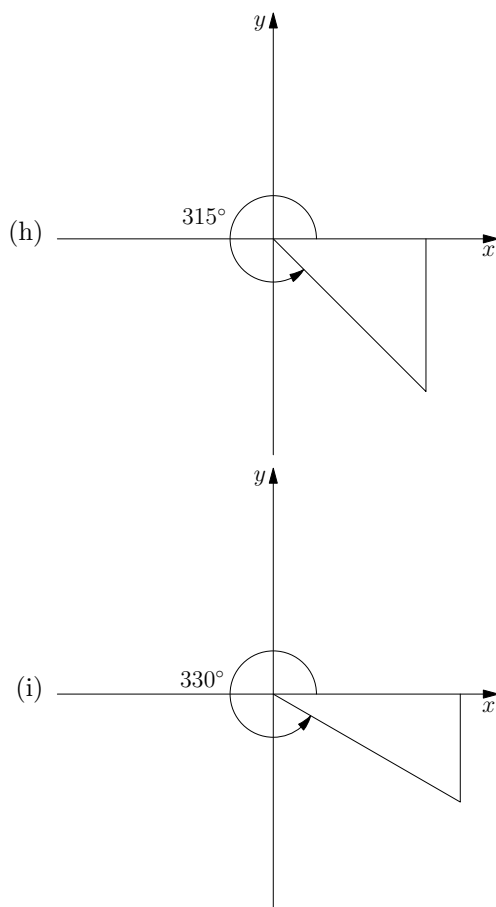
(h) $c = 1$

4. For each of the following diagrams find the angle in the triangle whose vertex is located at the origin of the coordinate system (the reference angle) and then label the side lengths of the triangle with the appropriate sign.









5. Find the following:

- | | | |
|----------------------|----------------------|----------------------|
| (a) $\sin 0^\circ$ | (g) $\sin 135^\circ$ | (m) $\sin 270^\circ$ |
| (b) $\sin 30^\circ$ | (h) $\sin 150^\circ$ | (n) $\sin 300^\circ$ |
| (c) $\sin 45^\circ$ | (i) $\sin 180^\circ$ | (o) $\sin 315^\circ$ |
| (d) $\sin 60^\circ$ | (j) $\sin 210^\circ$ | (p) $\sin 330^\circ$ |
| (e) $\sin 90^\circ$ | (k) $\sin 225^\circ$ | (q) $\sin 360^\circ$ |
| (f) $\sin 120^\circ$ | (l) $\sin 240^\circ$ | |

6. Graph each of the following

- | | | |
|---------------------|-------------------------------|-------------------------------|
| (a) $\sin \theta$ | (c) $5 \sin \theta$ | (e) $\frac{1}{4} \sin \theta$ |
| (b) $2 \sin \theta$ | (d) $\frac{1}{2} \sin \theta$ | (f) $.125 \sin \theta$ |

7. Find the following:

- | | | |
|----------------------|-----------------------|-----------------------|
| (a) $\sin 390^\circ$ | (d) $\sin 390^\circ$ | (g) $\sin 990^\circ$ |
| (b) $\sin 585^\circ$ | (e) $\sin 780^\circ$ | (h) $\sin 1035^\circ$ |
| (c) $\sin 480^\circ$ | (f) $\sin 1440^\circ$ | (i) $\sin 1620^\circ$ |

8. Graph each of the following

- | | | |
|--------------------|--------------------|------------------------------|
| (a) $\sin \theta$ | (d) $\sin 4\theta$ | (g) $\sin \frac{1}{2}\theta$ |
| (b) $\sin 2\theta$ | (e) $\sin 5\theta$ | (h) $\sin \frac{1}{4}\theta$ |
| (c) $\sin 3\theta$ | (f) $\sin 6\theta$ | |

Chapter 3

Vibration of strings

3.1 Frequency

When a string vibrates its motion repeats a pattern over and over. How many times that pattern repeats in a given time is called the frequency.

Definition 11 (Frequency). *The frequency of a repeating pattern is the number of times it occurs in a given time. The standard unit of measure is Hertz, Hz, which is cycles per second.*

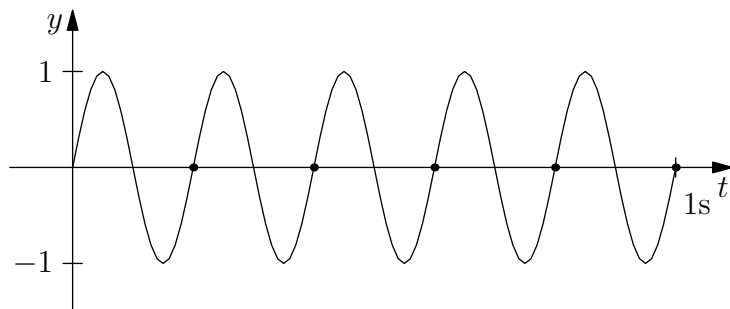


Figure 3.1: A sine wave with a frequency of 5 Hz (cycles/second).

We are surrounded by waves vibrating at many different frequencies. If you listen to the FM radio station “106.9 The Eagle” that 106.9 is the frequency at which that station is broadcasting. It is in fact 106.9 MHz where M stands for “mega” and means you are to multiply the 106.9 by 10^6 . Another prefix is k which stands for “kilo” and means to multiply by 10^3 .

While we are talking about radio, FM means frequency modulation and AM is amplitude modulation. These are two different analog methods of encoding sound and transmitting it using radio waves. FM radio uses the frequencies 87.5 MHz to 108.0 MHz while AM radio uses 535 kHz to 1700 kHz.

3.2 Fundamental Frequencies and Overtones

When a string is plucked (or bowed or hammered) it vibrates at specific frequency. In addition to this fundamental frequency, the string will also vibrate at higher frequencies. These higher frequencies are regular and in fact are integral multiples of the fundamental frequency. So if a string is vibrating at a frequency of f it will also be vibrating at $2f$, $3f$, $4f$, etc., see figure 3.2. These are called overtones or harmonics.

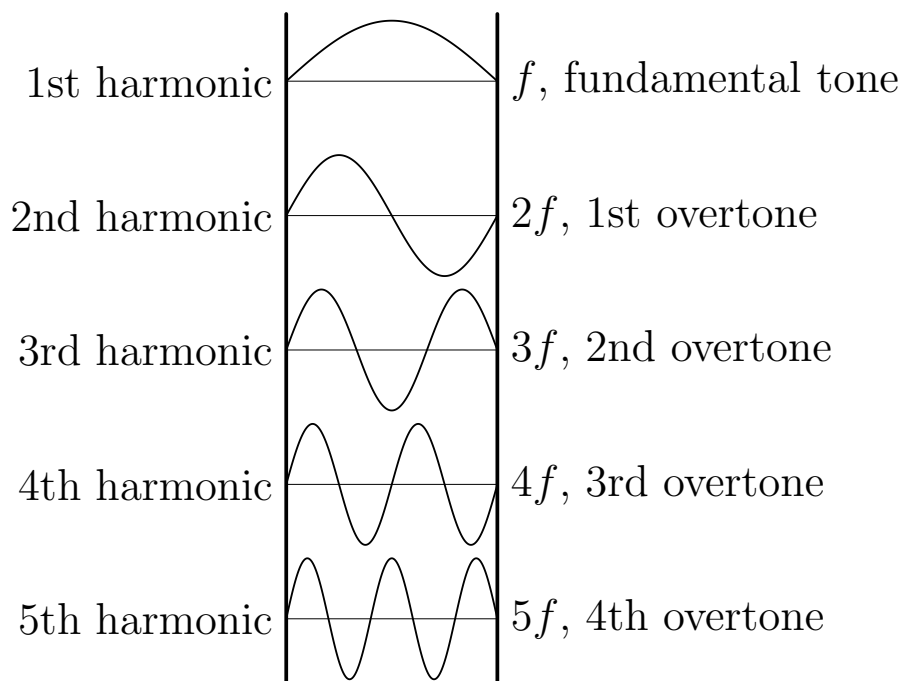


Figure 3.2: Fundamental tone, overtones, and harmonics.

The relationship between frequency, overtone, and harmonics can be seen in table 3.1.

Frequency	f	$2f$	$3f$	$4f$	$5f$	$\dots$
Overtones		1st	2nd	3rd	4th	$\dots$
Harmonics	1st	2nd	3rd	4th	5th	$\dots$

Table 3.1: Relationship between frequencies, overtones, and harmonics.

Example 3. For the fundamental frequency $f = 100$ Hz, let us find the 3rd overtone and the 5th harmonic.

Since the 3rd overtone is $4f$ we have $4 \cdot 100$ Hz = 400 Hz.

The 5th harmonic is $5f$ so it is $5 \cdot 100$ Hz = 500 Hz.

The collection of overtones (integral multiples of the fundamental series) is called a overtone series.

Example 4. Find the overtone series for the following frequencies:

$$1. f = 100\text{Hz} \qquad 2. f = 150\text{Hz} \qquad 3. f = 200\text{Hz}$$

Since the overtone series is just integral multiples of the fundamental frequency we just need to do the following multiplications:

$$1f, 2f, 3f, 4f, 5f, 6f, \dots$$

1. When $f = 100$ Hz the overtone series is

$$100 \text{ Hz}, 200 \text{ Hz}, 300 \text{ Hz}, 400 \text{ Hz}, 500 \text{ Hz}, 600 \text{ Hz}, \dots$$

2. When $f = 150$ Hz the overtone series is

$$150 \text{ Hz}, 300 \text{ Hz}, 450 \text{ Hz}, 600 \text{ Hz}, 750 \text{ Hz}, 900 \text{ Hz}, \dots$$

3. When $f = 200$ Hz the overtone series is

$$200 \text{ Hz}, 400 \text{ Hz}, 600 \text{ Hz}, 800 \text{ Hz}, 1000 \text{ Hz}, 1200 \text{ Hz}, \dots$$

3.3 Fourier series and Timbre

Suppose we were to add together the following list of sine functions:

$$\sin 100t, \sin 200t, \sin 300t, \sin 500t, \sin 600t, \dots,$$

then what we would get is called a Fourier series:

$$\sin 100t + \sin 200t + \sin 300t + \sin 500t + \sin 600t + \dots$$

Usually the sine functions have a constant in front that changes the amplitude so our series could look more like:

$$F = 2 \sin 100t + 5 \sin 200t + 3 \sin 300t + 12 \sin 500t + 6 \sin 600t + \dots$$

Let us make the following definition:

Definition 12 (Fourier series). A Fourier series is an infinite sum of sine functions whose frequencies are integral multiples of the fundamental frequency

f and can be written as:¹

$$F = a_1 \sin ft + a_2 \sin 2ft + a_3 \sin 3ft + a_4 \sin 4ft + \cdots$$

Some of these constants: $a_1, a_2, a_3, \dots$, may be zero.

In our class there will only be a finite number of non-zero terms.

Example 5. Let us write the Fourier series for the wave with frequencies 10 Hz, 20 Hz, and 30 Hz with respective amplitudes 10, 3, and 6.

Since there are only three frequencies our Fourier series will only have three terms:

$$F = 10 \sin 10t + 3 \sin 20t + 6 \sin 30t.$$

3.3.1 Timbre

Recall from the previous section that anytime we play a tone on an instrument we get the fundamental frequency and the overtones above it. From a mathematical point of view we get an infinite number of overtones, but in the physical world only a finite number of them can be produced.

If a piano and a guitar are both playing the same note, how do we tell them apart? In music the property that allows us to do this is called timbre (pronounced, using IPA symbols, /¹tæmbər/ or my best approximation “tamber”).

¹This is not strictly true. A Fourier series typically also contains cosine functions as well as a constant term:

$$F = a_0 + a_1 \sin ft + b_1 \cos ft + a_2 \sin 2ft + b_2 \cos 2ft + a_3 \sin 3ft + b_3 \cos 3ft + \cdots$$

or written in a more compact form

$$F = a_0 + \sum_{n=1}^{\infty} (a_n \sin nft + b_n \cos nft)$$

We are not going to worry about these details since they will not help in understanding the phenomenon we are investigating.

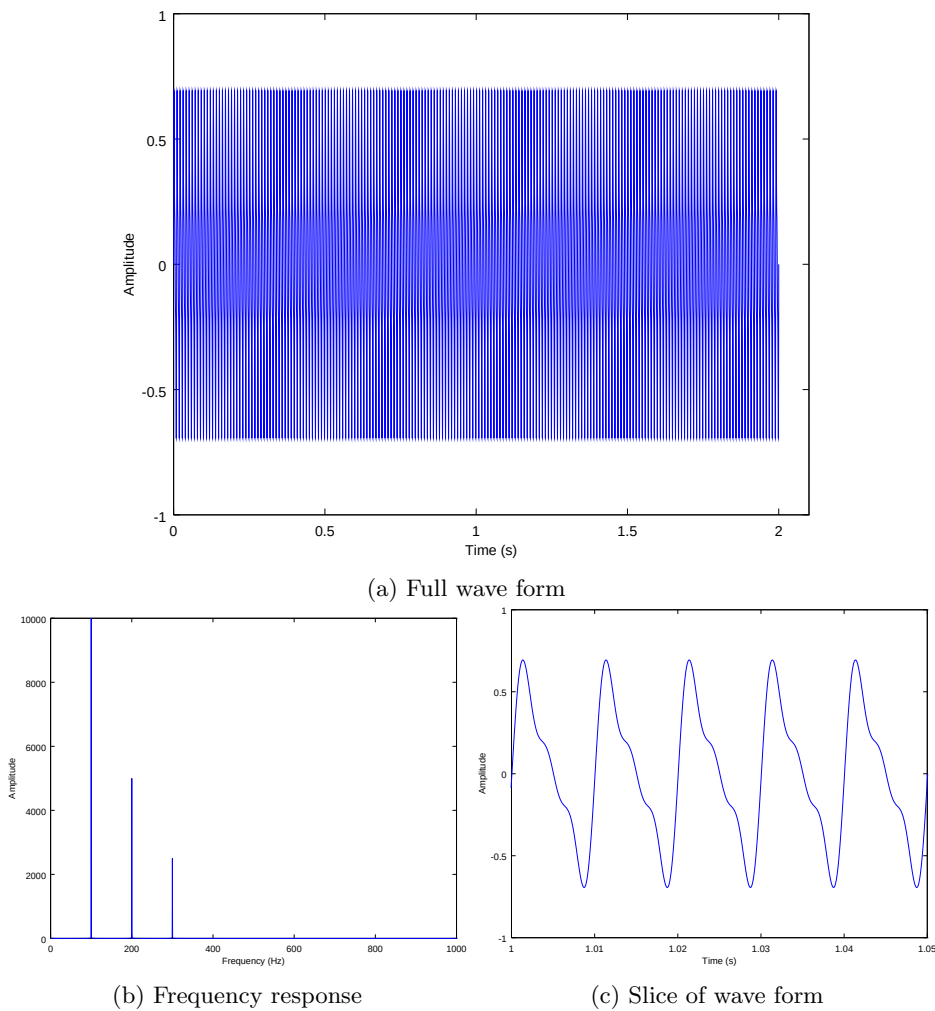


Figure 3.3: A tone consisting of a 100 Hz fundamental frequency and overtones of 200 Hz and 300 Hz.

Suppose we have a tone that consists of a fundamental frequency of 100 Hz and overtones of 200 Hz and 300 Hz with respective amplitudes of 10000, 5000, and 2500. Then the Fourier series² will be

$$F = 10000 \sin 100t + 5000 \sin 200t + 2500 \sin 300t.$$

The graph of this sound wave played for 2 s can be seen in figure 3.3a. In figure 3.3c we can see approximately 5 cycles of this wave.

²These amplitudes are not accurate in terms of magnitude, but their distribution is a good enough representation of reality for our needs that we will not worry about it.

Now suppose we have the sound wave represented in figure 3.3a and we would like to know what frequencies and at what amplitudes are the sine waves that make it up. There is a mathematical tool that can take a sound wave and separate out the frequencies and their amplitudes. It is called a Fourier transform and the details of how it works is well beyond the scope of this class. But all we need to know is what it does and that we can use it to generate a “Frequency response” graph such as the one shown in figure 3.3b which shows the frequencies and their amplitudes.

We are now ready to discuss timbre.

Definition 13 (Timbre). *Timbre is the set of frequencies and their respective amplitudes that an instrument produces.*

If a piano and a guitar play the same note the reason we can tell them apart is because they will have different amplitudes for the same frequencies.

Below we have the graph of the sound wave, frequency response graph, and a slice of the sound wave of five instruments all playing an A4, i.e. a tone at 440 Hz. The instruments and there respective figures are flute 3.4, piano 3.5, singer 3.6, tuning fork 3.7, and violin 3.8. The “.wav” files can be found at <http://mcis.jsu.edu/faculty/desmith/ms108/music/music.html> which I found at <http://www.johnloomis.org/ece303L/notes/music/wavfiles/A4.html>.

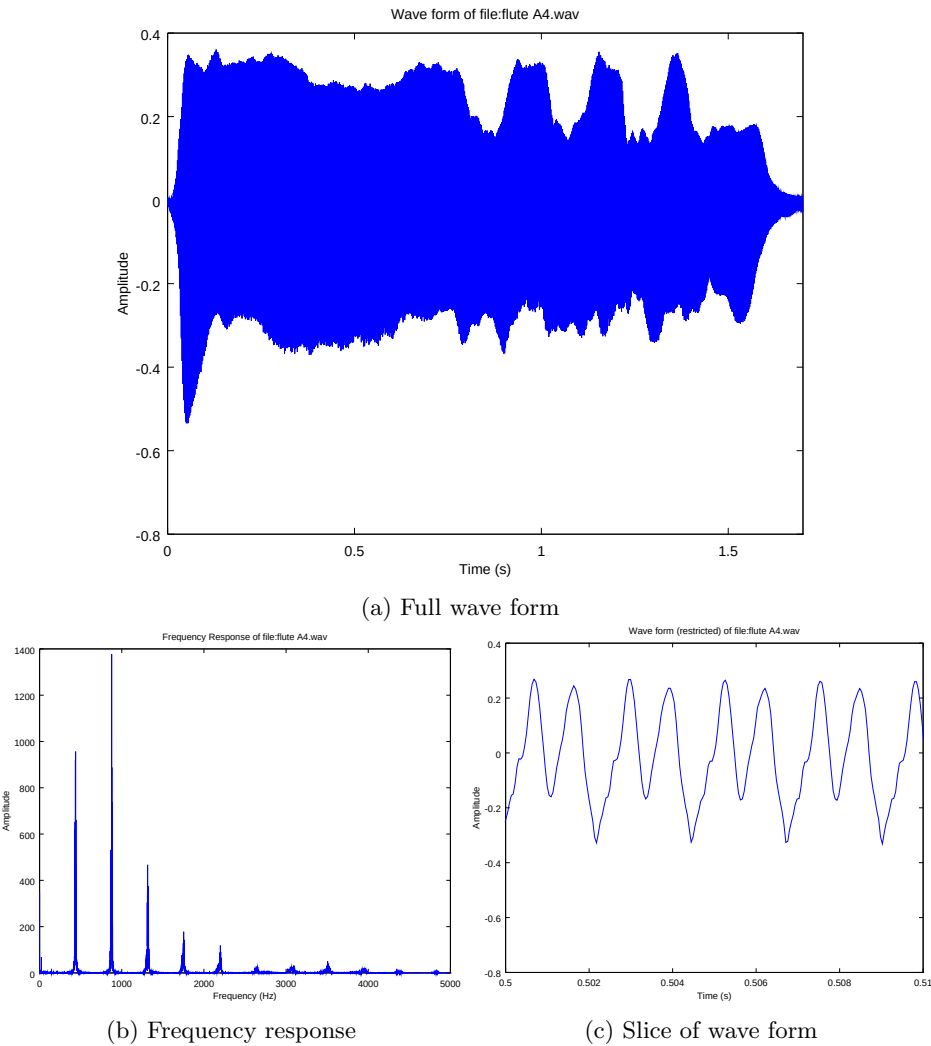
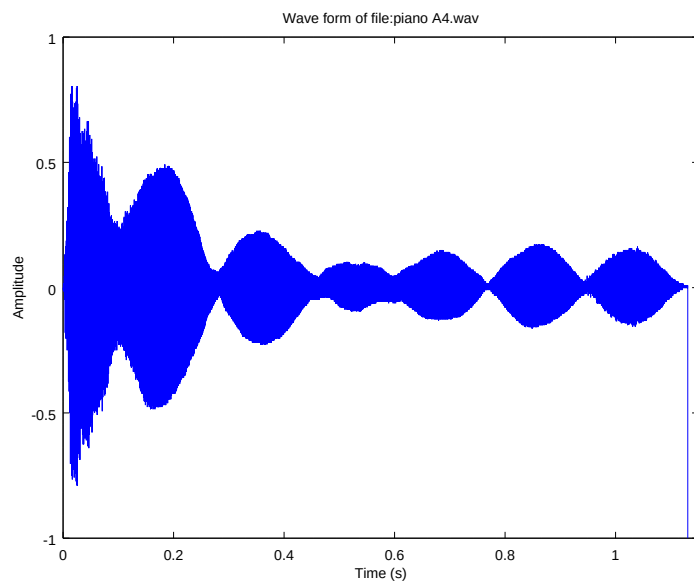
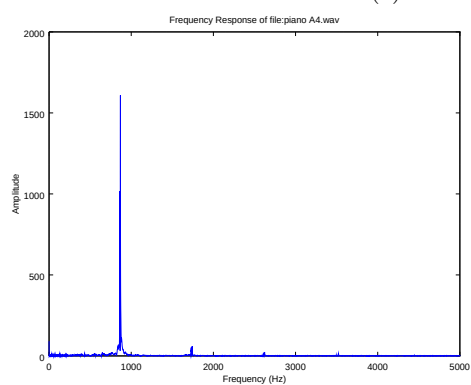


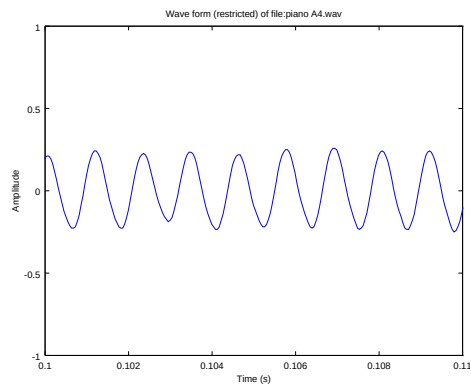
Figure 3.4: Flute playing an A4



(a) Full wave form



(b) Frequency response



(c) Slice of wave form

Figure 3.5: Piano playing an A4

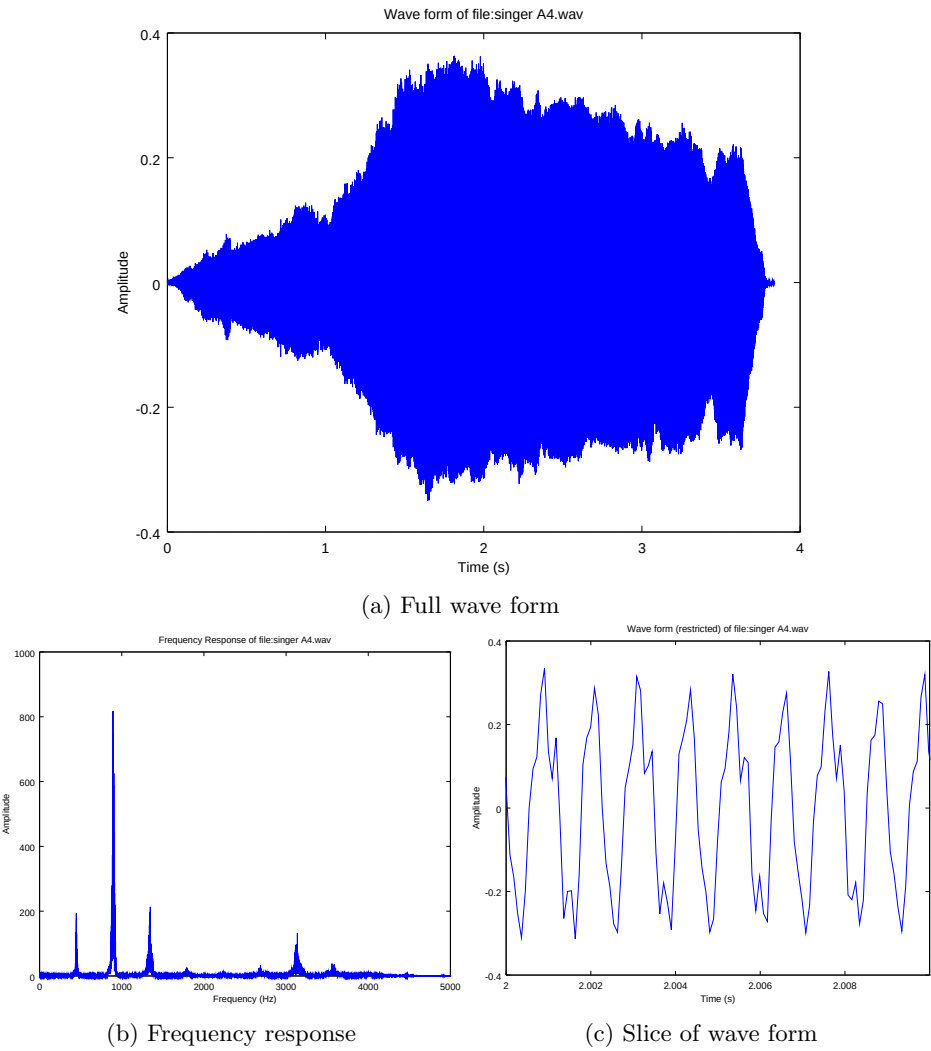
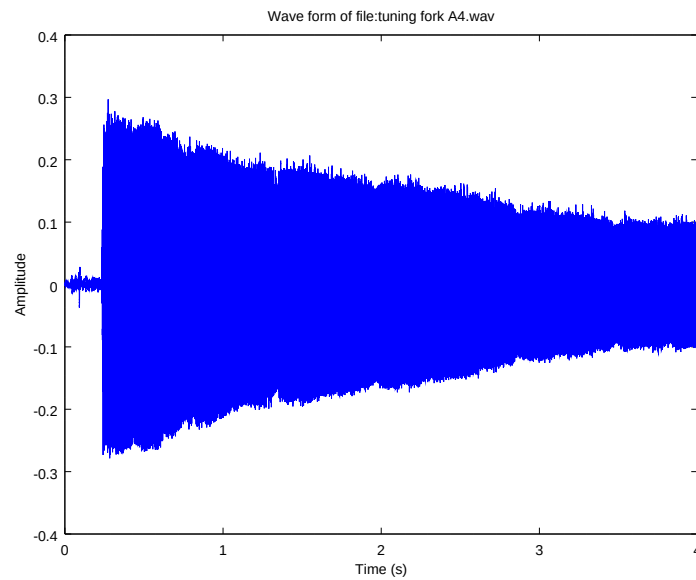
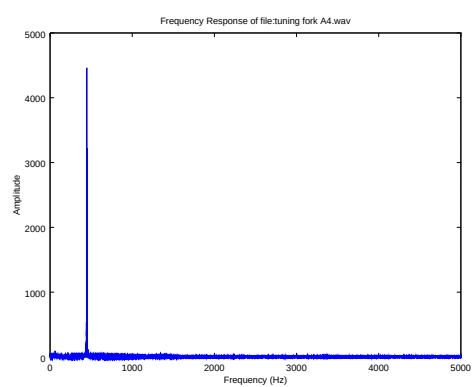


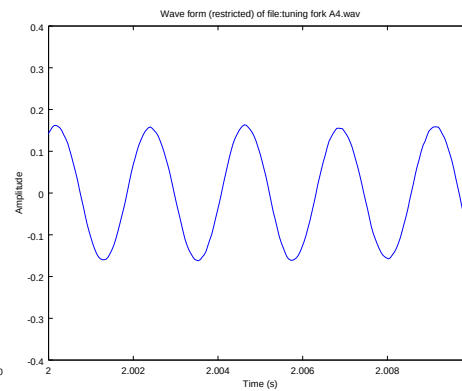
Figure 3.6: Singer singing an A4



(a) Full wave form



(b) Frequency response



(c) Slice of wave form

Figure 3.7: Tuning fork playing an A4

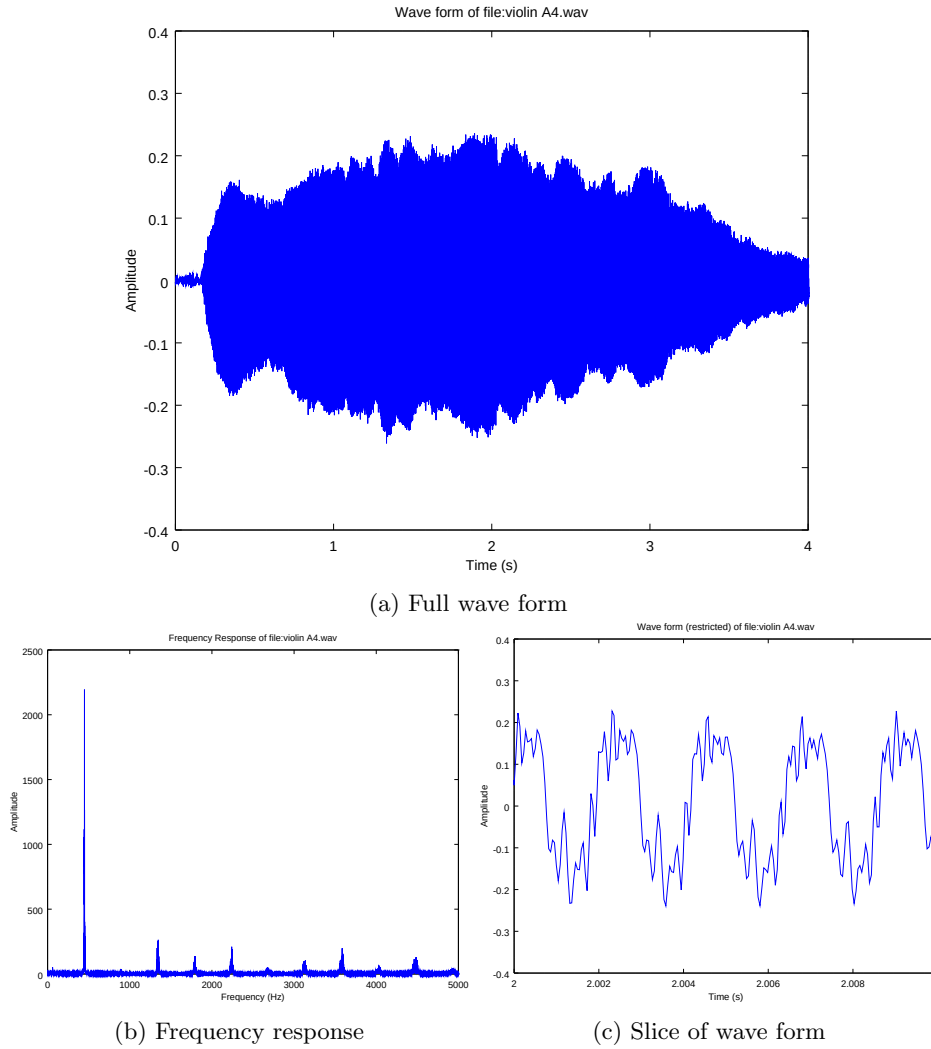


Figure 3.8: Violin playing an A4

We can see that they all consists of multiplies of 440 Hz but with various amplitudes.

3.4 String frequency formula

The fundamental frequency, f , of a string is given by the formula:

$$f = \frac{1}{2L} \sqrt{\frac{T}{\rho}}$$

where L is the string length, T is the tension, and ρ is the density. ???NEED TO ADD AND NEED MORE TIME???

3.5 Homework

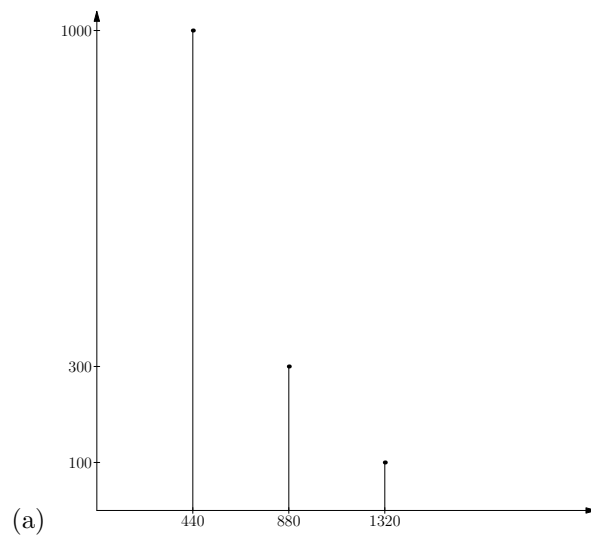
1. What is the relationship between harmonics and overtones?
2. Calculate the first five harmonics for the following fundamental frequencies.

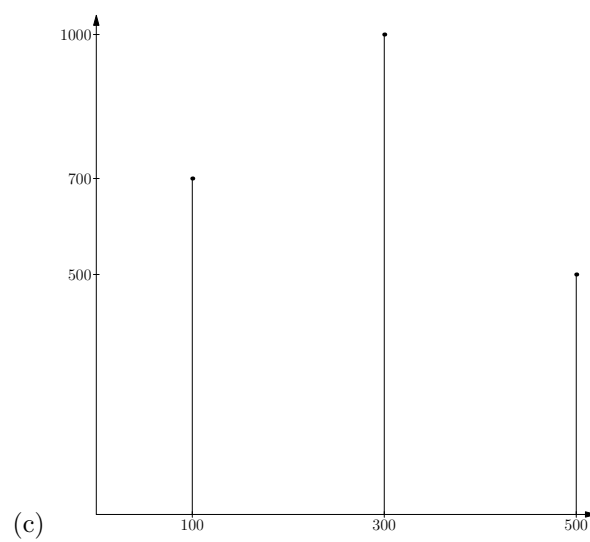
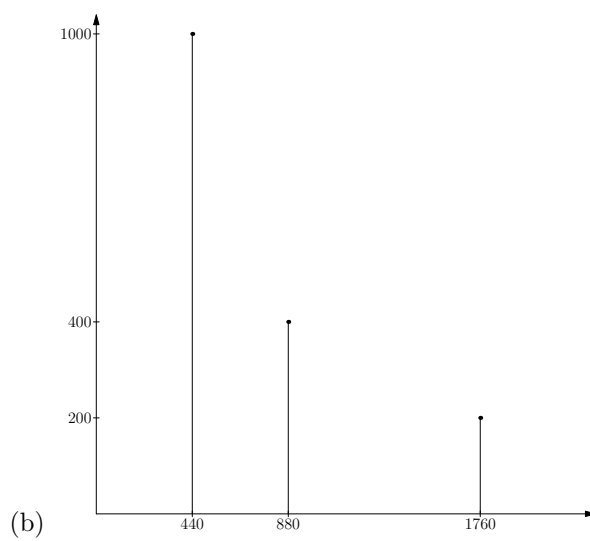
- | | | |
|------------|------------|------------|
| (a) 100 Hz | (c) 440 Hz | (e) 50 Hz |
| (b) 200 Hz | (d) 20 Hz | (f) 436 Hz |

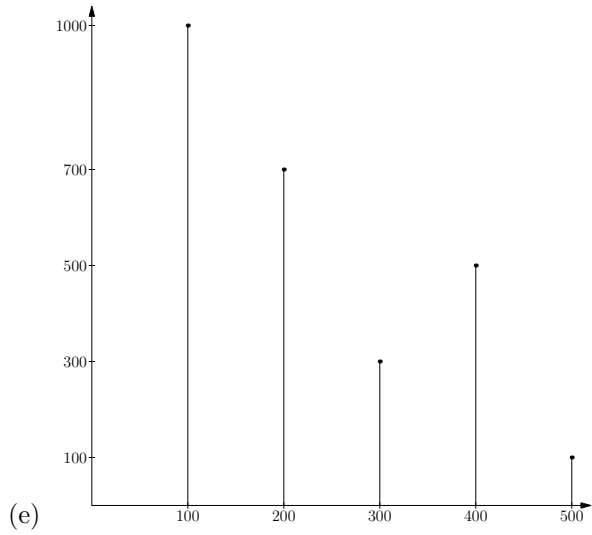
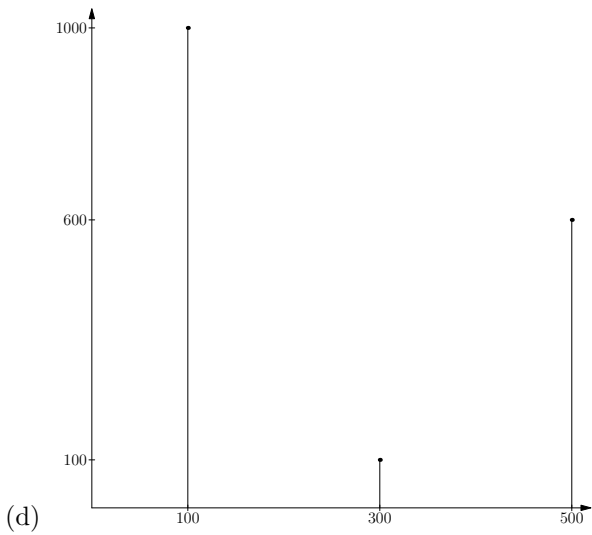
3. Find the overtone series for the following frequencies (give at least five).

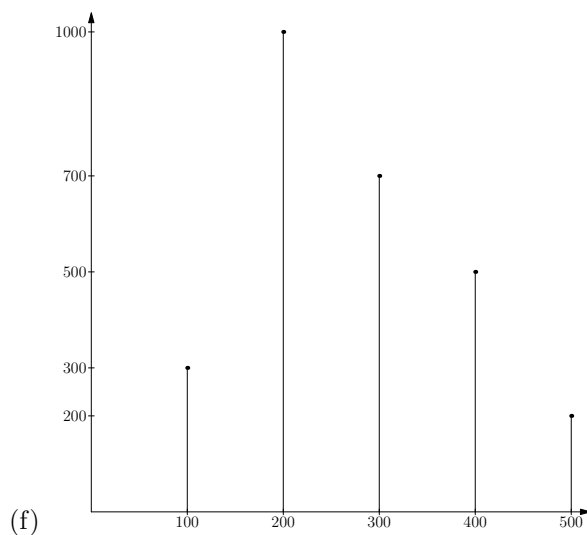
- | | | |
|-----------|------------|-------------|
| (a) 10 Hz | (d) 100 Hz | (g) 1 Hz |
| (b) 15 Hz | (e) 150 Hz | (h) 27.5 Hz |
| (c) 20 Hz | (f) 200 Hz | (i) 440 Hz |

4. For the following frequency response graphs find the Fourier series.









5. For the following Fourier series make the frequency response graphs.

- (a) $F = 1000 \sin(440t) + 300 \sin(880t) + 200 \sin(1320t)$
- (b) $F = 1000 \sin(50t) + 700 \sin(100t) + 400 \sin(150t)$
- (c) $F = 600 \sin(50t) + 1000 \sin(100t) + 300 \sin(200t)$
- (d) $F = 500 \sin(100t) + 1000 \sin(200t) + 600 \sin(300t) + 3 \sin(400t)$

Chapter 4

Music

We are finally getting to the musical portion of the notes. Suppose we wish to make some music. Since music consists of sounds we need to figure out how to produce sound waves at various frequencies. This has been solved many times over by every culture and has resulted in the great variety of instruments we are familiar with and possibly many we are not. So bass guitars, pianos, saxophones, flutes, bagpipes, and the human voice are among many of the instruments we probably are familiar with. A list of instruments can be found at https://en.wikipedia.org/wiki/List_of_musical_instruments.

We can produce any frequency we want within reason, but not all frequencies sound good together. That is what we want to investigate next.

4.1 Diatonic and Chromatic scales

Notes are often grouped together in an ascending collection called a scale. For the time being we are going to concentrate on a scale that consists of seven notes. This is called a diatonic, or heptatonic (*hepta* meaning seven), scale. We will use the first seven letters from our alphabet to denote these tones: *A*, *B*, *C*, *D*, *E*, *F*, *G*. But for reasons that we won't go into here¹ this is not the order that we will usually write them in. Instead we will start with *C*, go in order, and then stop also at *C*. This is called a *C* major scale and written CM:

C, *D*, *E*, *F*, *G*, *A*, *B*, *C*.

To add confusion to this, the note most often used as a reference note is not *C*, but is instead *A*. We will come back to this in a moment.

We will not use this in our class, but you can also name the tones using the Solfège method. This consists of the syllables (if you have seen the *The Sound of Music* you will recall this):

¹I lied, historically the Aeolian church mode (*A*, *B*, *C*, *D*, *E*, *F*, *G*) was the dominant mode and the first tone in that scale was named *A*. Later the Ionian mode later became dominant and it starts with the tone that was already called a *C* in the Aeolian mode. The more common names for these two modes are Major (Ionian mode) and minor (Aeolian).

Do, Re, Me, Fa, Sol, La, Ti, Do.

but these do not correspond to fixed notes². For this reason among others we will stick with the *letter notation*.

Definition 14 (Letter notation). *Letter notation is a system for designating the different tones with the first seven letters of the Roman alphabet: A, B, C, D, E, F, G. Tones can be shifted by a up by a semi-tone using a sharp $\sharp$ or down a semi-tone using a flat $\flat$.*

Don't worry about the sharps and flats for now, they are included in the definition for completeness.

The *C* major scale corresponds to the white keys on a modern piano, figure 4.1.

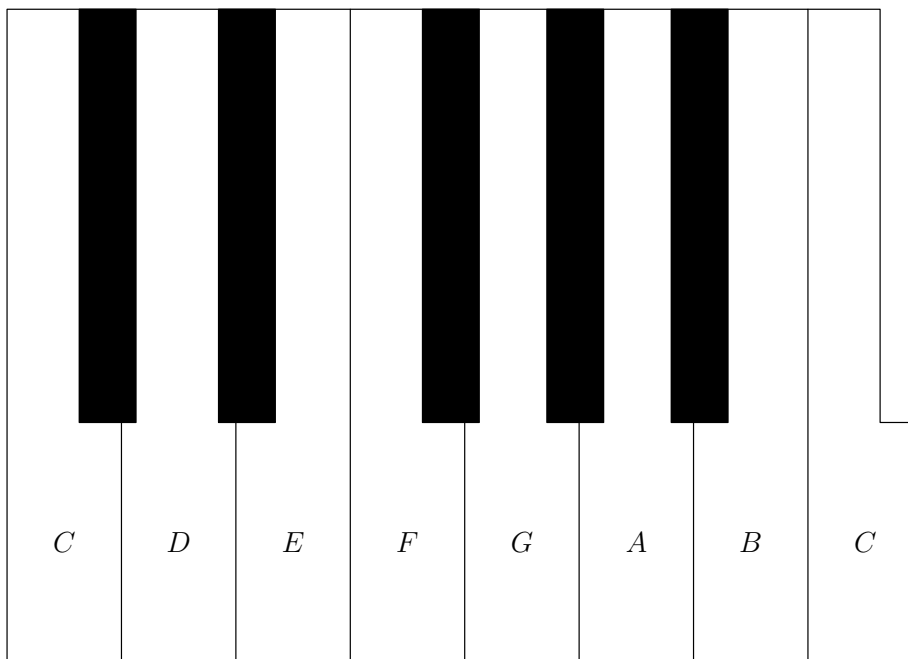


Figure 4.1: Piano key diagram with diatonic scale.

What frequencies should we associate with these note names? It turns out that the note called *A* could have been chosen to be any frequency, but once it was chosen the other notes are in a certain sense fixed. The standard reference note is called *A4* and has a frequency of 440 Hz. The frequencies of the other notes in relation to *A4* is called a tuning system.

²Unless you are using the Fixed Do solfège in which case it does. In countries that use this method *C* and Do are equivalent (according to https://en.wikipedia.org/wiki/Solf%C3%A8ge#Fixed_do_solf.C3.A8ge).

Definition 15 (Tuning system). *A tuning system is the collection of frequencies assigned to the different notes.*

4.1.1 A mathematical interlude³

Ratios are a crucial to understanding the mathematics of music, so we are going to take a moment to discuss them. A good resource for this is Khan academy: <https://www.khanacademy.org/> Sign in (it's free) and search for "ratios". If you would like more homework on ratios they can be found here (this is also free): <https://www.knewton.com/login?r=%2Flearn>

A key fact to remember is that there two notations for ratios and we will jump between without much comment. The ratio of a to b can be written in the two equivalent forms:

$$a : b \text{ is the same as } \frac{a}{b}.$$

4.1.2 Pythagorean tuning and Just intonation

The mathematician Pythagoras⁴ is said to have been the first to study the relationship between tones and ratios of lengths of vibrating strings that produce those notes. We need some terminology to express how far apart two notes are, so let us make the following definition:

Definition 16 (Interval). *An interval is the distance between two tones and is the ratio of the frequencies of those two notes.*

Recall that the C Major scale we looked at in the beginning section 4.1 is

$$C, D, E, F, G, A, B, C.$$

The second C in the above list is said to be one octave higher than the first C . The ratio of frequencies of these two notes is $2 : 1$, as a fraction $(2/1)$, and this is true for any two notes an octave apart. So we can say that the interval from the first C to the second is one octave.

The story goes that Pythagoras built a monochord⁵ and as he changed the length of the string using integer ratios he observed that certain ratios corresponded to tones that sounded pleasing with each other, i. e. that they were consonant.

Definition 17 (Consonance/Dissonance). *Two tones are said to be consonant if they sound good when played together and dissonant if they do not sound good when played together.*

So what interval is most consonant with a given tone? It is an interval with the ratio of $1 : 1$, i.e. the note that sounds best with a given note is itself and is

³ Which I stole from someone, but I can't remember who.

⁴<https://en.wikipedia.org/wiki/Pythagoras>

⁵<https://en.wikipedia.org/wiki/Monochord>

called the unison. The second most consonant interval is the ratio of an octave, $2 : 1$. The next most consonant interval is the ratio called a fifth, $3 : 2$. In table 4.1 we have a diatonic scale (seven notes plus the octave) using a tuning called a Pythagorean tuning.

Tone	C	D	E	F	G	A	B	C
Scale degree	Unison	Second	Third	Fourth	Fifth	Sixth	Seventh	Octave
Rank by consonance	1	7	5	4	3	6	8	2
Ratio	$1/1$	$9/8$	$81/64$	$4/3$	$3/2$	$27/16$	$243/128$	$2/1$
Ratio as powers	$3^0/2^0$	$3^2/2^3$	$3^4/2^6$	$2^2/3^1$	$3^1/2^1$	$3^3/2^4$	$3^5/2^7$	$2/3^0$

Table 4.1: Pythagorean tuning

A tuning is called a Pythagorean tuning if the intervals are Pythagorean intervals:

Definition 18 (Pythagorean interval). *An interval is a Pythagorean interval if it is a ratio of 2 and 3 raised to some power.*

And so we can define a Pythagorean tuning as:

Definition 19 (Pythagorean tuning). *A Pythagorean tuning is a tuning system that only uses Pythagorean intervals, ratios of powers of 2 and 3.*

We can construct these ratios using octaves, fourths, and fifths by following the procedure⁶ below and shown in table 4.2:

1. Find the ratio for C , the fourth above it, F , and the fifth above it, G .
2. Since we have to preserve octaves we can fill in the ratios for C , F , and G one octave higher.
3. Next add the fifth above G which is a D and the D one octave down.
4. Now we find the perfect fifths above the D 's which are the A 's.
5. Then the perfect fifths above the A 's which are the E 's.
6. Finally the perfect fifths above the E 's which are the B 's.

Step	C	D	E	F	G	A	B	C	D	E	F	G	A	B
1	1			$\frac{4}{3}$	$\frac{3}{2}$									
2								2			$\frac{8}{3}$	3		
3		$\frac{9}{8}$							$\frac{9}{4}$					
4						$\frac{27}{16}$							$\frac{27}{8}$	
5			$\frac{81}{64}$							$\frac{81}{32}$				
6							$\frac{243}{128}$							$\frac{243}{64}$

Table 4.2: Construction of the Pythagorean tuning

⁶Follows the construction in Leon Harkleroad's The Math behind the Music pgs 23–25.

Let us look at why certain intervals sound better with the unison than others do. We will compare the overtone series for the unison, octave, fifth, and seventh using a frequency of $f = 100$ Hz, table 4.3. If we compare the overtone series for the octave to the overtone series for the unison we can see that every frequency in the octave is also a frequency for the unison. This is why an octave sounds like the same as tone as the note one octave above or below it.

Next let us look at the fifth. It has a ratio of $3 : 2$ and so it has a frequency that is the second overtone, $3f$, dropped by one octave, divided by 2. When we look at the fifth's overtone series we see that its frequencies correspond to every other frequency in the unison's overtone series. This means that the physical sound waves are almost never interfering with each other relatively speaking.

Finally, we look at the overtone series for the interval of a seventh ($243 : 128$) using the Pythagorean tuning. It's frequencies and the unison's frequencies never align in the table. It will take the 127th overtone in the seventh's overtone series to align with the unison. This will be at a frequency of 24300 Hz which is above the range at which humans can hear, approximately 20 Hz to 20 kHz. This is supposing that the instrument could even produce this overtone in the audible range of humans. Recall that as we get higher in the overtone series those frequencies generally have decreasing amplitudes, meaning that the volume is decreasing. So the physical sound waves of the seventh and unison never work together and we hear this as dissonance.

The Pythagorean tuning works well if you mainly use fifths and fourths.

If we look back at table 4.1 we will see that some of the ratios consist of large integers. What if we try to use only “small” integers in the ratios that make up the tuning system? This leads to:

Definition 20 (Just intonation). *A just intonation is a tuning system that only uses ratios of small integers.*

An example of a just intonation can be seen in the table 4.4 below:

Unison	Octave	Fifth	Seventh
100			
		150	
			189.84
200	200		
300		300	
			379.69
400	400		
		450	
500			
			569.53
600	600	600	
700			
		750	
			759.38
800	800		
900		900	
			949.22
1000	1000		
		1050	
1100			
			1139.06
1200	1200	1200	
1300			
			1328.91
		1350	
1400	1400		
1500		1500	
			1518.75
1600	1600		
		1650	
1700			
			1708.59
1800	1800	1800	
			1898.44
1900			
		1950	
2000	2000		
			2088.28

Table 4.3: Overtone series for Unison, Octave, Fifth, and Seventh using $f = 100$ Hz and Pythagorean tuning.

Tone	C	D	E	F	G	A	B	C
Scale degree	Unison	Second	Third	Fourth	Fifth	Sixth	Seventh	Octave
Rank by consonance	1	7	6	4	3	5	8	2
Ratio	1/1	9/8	5/4	4/3	3/2	5/3	15/8	2/1

Table 4.4: Just Intonation

Note that it only differs from the Pythagorean tuning above in the intervals of a third, sixth, and seventh. These smaller integer ratios result in a better sounding scale. ???WHY? NEED TO ADD MORE LATER???

Let us construct the Just intonation ratios by following the procedure⁷ below and shown in table 4.5:

1. We start with the ratio for C and use the small integer ratios from the overtone series to get the fifth G , the fourth F , the third E , and the sixth A .
2. Going up an octave for each we get the next line.
3. A fifth above an E is the B , so we do that is next and it's octave.
4. The final note is the D which we will make a fifth above G and then drop down an octave for the other D .

Step	C	D	E	F	G	A	B	C	D	E	F	G	A	B
1	1		$\frac{5}{4}$	$\frac{4}{3}$	$\frac{3}{2}$	$\frac{5}{3}$								
2								2		$\frac{5}{2}$	$\frac{8}{3}$	3	$\frac{10}{3}$	
3							$\frac{15}{8}$							$\frac{15}{4}$
4		$\frac{9}{8}$							$\frac{9}{4}$					

Table 4.5: Construction of Just intonation

Note that both the Pythagorean tuning and just intonation are based on the importance of the fifth.

???SHOW THAT THE INTERVAL BETWEEN SCALE DEGREES ARE NOT THE SAME???

For reasons we don't have time to discuss now modern Western music uses 12 tones per octave. We have already seen 7 of them: A, B, C, D, E, F , and G . These are the white keys on a piano. Between some of these notes we have additional notes which are the black keys on the piano. So between C and D , D and E , F and G , G and A , and A and B .

Instead of having to rename the notes musicians added a symbol to those notes to symbolize these new notes. In order to make our discussion clear let us introduce a definition at this time.

⁷Again following the construction in Leon Harkleroad's The Math behind the Music pgs 26–27.

Definition 21 (Whole step/half-step or whole tone/semitone). A *half-step* is the distance between any two adjacent keys on a piano. A *whole step* consists of two half steps. A half step is sometimes called a *semitone* and a whole step is a *whole tone*.

Definition 22 (Sharp ($\sharp$), Flat ($\flat$)). A *sharped note* is a note a half-step above and is denoted by adding the symbol $\sharp$ to it. A *flattened note* is a note a half-step below and is denoted by adding the symbol $\flat$ to it.

So the distance between an E and an F is a half-step as is the distance from a B to a C . If we go up a half-step from C we get a $C\sharp$. A half-step above a D is a $D\sharp$. But a half-step above an E is a F ⁸, see figure 4.2.

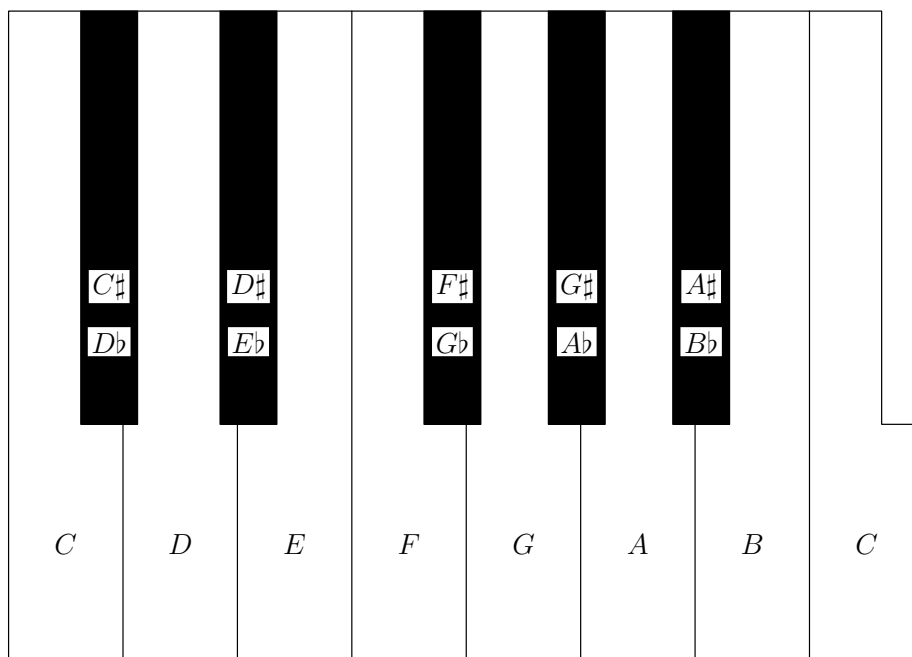


Figure 4.2: Piano key diagram with chromatic scale.

If we go a half-step down from a D we get $D\flat$, a half-step down from a G we get $G\flat$, but a half-step down from a F is an E because they are adjacent on the piano keyboard.

A fifth above C is still a G , but if we now count, starting with $C\sharp$, using semi-tones it is 7 semitones above the C . A fourth is 5 semitones above the C and is still an F .

If we take consecutive fifths starting with F we will get the following list of

⁸This means that an $E\sharp$ is an F and a $B\sharp$ is a C . It also means that a $F\flat$ is an E and a $C\flat$ is a B

notes:

$$F, C, G, D, A, E, B, G\flat, D\flat, A\flat, E\flat, B\flat$$

and the fifth above $B\flat$ is an F . If we continue taking fifths will will just repeat this pattern. This results in the circle of fifths, figure 4.3:

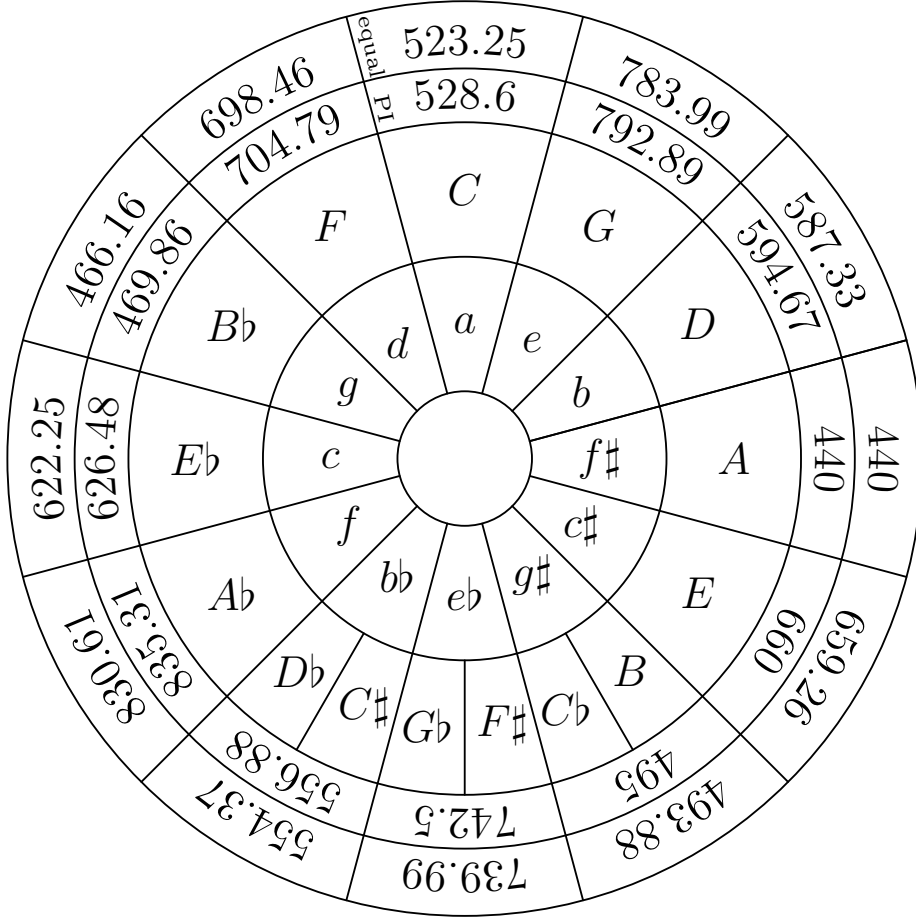


Figure 4.3: Circle of fifths with Pythagorean and equal temperament intervals starting with A440.

So starting with any note if we go twelve fifths above it we will get the same note, just with a higher pitch. Or do we? Recall that we can find any fifth above our reference frequency, f , using the formula

$$f_{n\text{-fifth}} = f \cdot \left(\frac{3}{2}\right)^n$$

and the any octave above our reference frequency, f , using the formula

$$f_{n\text{-oct}} = f \cdot (2)^n.$$

According to the circle of fifths the twelfth fifth should be, it turns out, the seventh octave. So we should have the following relationship $f_{12.\text{fifth}} = f_{7.\text{oct}}$ and therefore

$$\begin{aligned} f \cdot \left(\frac{3}{2}\right)^{12} &= f \cdot 2^7 \\ \left(\frac{3}{2}\right)^{12} &= 2^7 \\ 129.74 &\neq 128 \end{aligned}$$

As we can see they are not equal. This causes problems.
 ???ADD STUFF LATER???

4.1.3 Equal temperament

We cannot alter the octave. it must always be a ratio of 2 : 1 going up or 1 : 2 going down. Instead we will alter the fifth. But if we do not set the fifth to a ratio of 3 : 2 what should it be? To determine this we will solve the following equation

$$r^{12} = 2^7.$$

To do this we will take the twelfth root of both sides to get

$$r = \sqrt[12]{2^7} = 2^{\frac{7}{12}} \approx 1.498.$$

Note that this is close to the ratio of $\frac{3}{2} = 1.5$ which is the value of a fifth we found using the overtone series.

Recall that a fifth is 7 semitones above the tonic, we have 12 semitones per octave, and we are now defining a fifth to be $2^{\frac{7}{12}}$. It is therefore reasonable to let the distance between semitones be given by $2^{\frac{1}{12}}$. Since the intervals between any two consecutive semitones are now the same this tuning system is called the *equal temperament* tuning.

4.2 Beats, not the rhythmic kind

To see what happens when we play two frequencies at the same time we need the following identity from trigonometry:

Identity 1 (Sum-to-Product for sine).

$$\sin a + \sin b = 2 \sin \left(\frac{a+b}{2} \right) \cos \left(\frac{a-b}{2} \right)$$

Suppose we play two notes with frequencies very close to one another, say $f_1 = 102$ Hz and $f_2 = 100$ Hz. What will be the result? From the identity above we have:

$$\sin 102t + \sin 100t = 2 \sin \left(\frac{102+100}{2}t \right) \cos \left(\frac{102-100}{2}t \right) = 2 \sin 101t \cos t.$$

What this looks like is shown below:

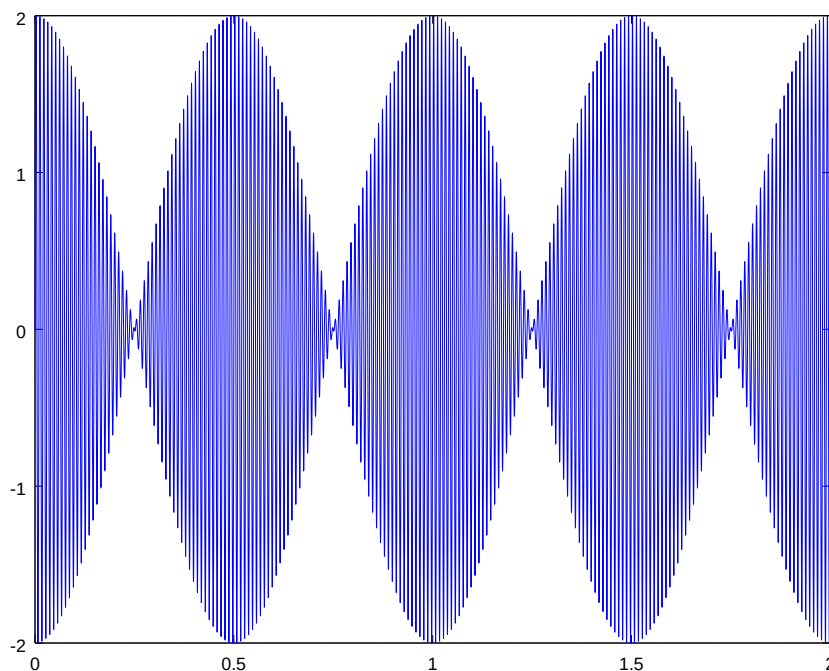


Figure 4.4: The wave form when tone of 102 Hz and 100 Hz are played together.

Since $\sin 102t + \sin 100t = 2 \sin 101t \cos t$ what we hear are two different effects. The first is a note with a frequency of 101 Hz and this comes from the $\sin 101t$ factor. The second is the volume of this sound will go from its maximum to 0 and back to its maximum twice in the span of one second due to the $\cos t$ factor. We will say that the beat rate is $\frac{1}{2}$ Hz.

If we combine tones with frequencies of $f_1 = 101$ Hz and $f_2 = 100$ Hz we get the following:

$$\sin 101t + \sin 100t = 2 \sin \left(\frac{101 + 100}{2} t \right) \cos \left(\frac{101 - 100}{2} t \right) = 2 \sin 100.5t \cos \frac{1}{2}t.$$

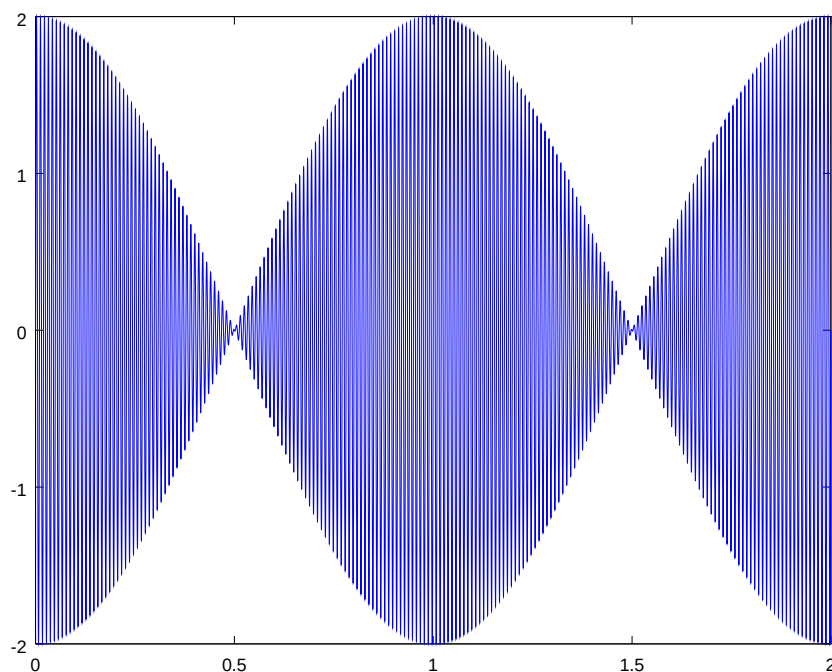


Figure 4.5: The wave form when tone of 101 Hz and 100 Hz are played together.

So the frequency is 100.5 Hz. The beat rate is 1 Hz here. The way we calculate it is to do the following, take the frequency from the cosine factor, $\frac{1}{2}$ and find its reciprocal 2. The beat rate is half of the reciprocal so $\frac{1}{2} \cdot 2 = 1$ Hz.

Example 6. Find the frequency and beat rate for the following frequencies:

1. $f_1 = 50.5$ Hz and $f_2 = 50$ Hz
2. $f_1 = 50.1$ Hz and $f_2 = 50$ Hz

1. First we calculate

$$\sin 50.5t + \sin 50t = 2 \sin \left(\frac{50.5 + 50}{2} t \right) \cos \left(\frac{50.5 - 50}{2} t \right) = 2 \sin 50.25t \cos 0.25t.$$

So the frequency is 50.25 Hz. To find the beat rate we take 0.25 and find its reciprocal which is 4. Now halve 4 to get a beat rate of $\frac{1}{2} \cdot 4 = 2$ Hz.

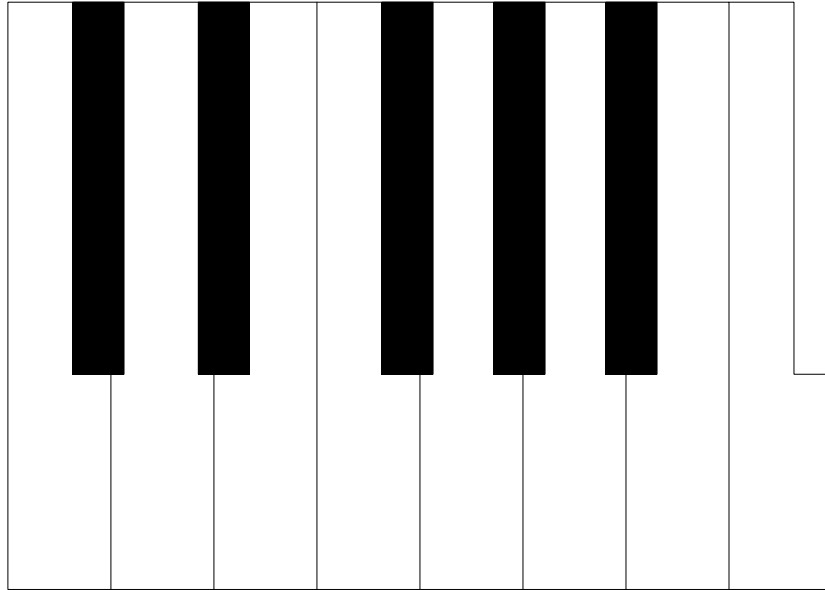
2. First we calculate

$$\sin 50.1t + \sin 50t = 2 \sin \left(\frac{50.1 + 50}{2} t \right) \cos \left(\frac{50.1 - 50}{2} t \right) = 2 \sin 50.05t \cos 0.05t.$$

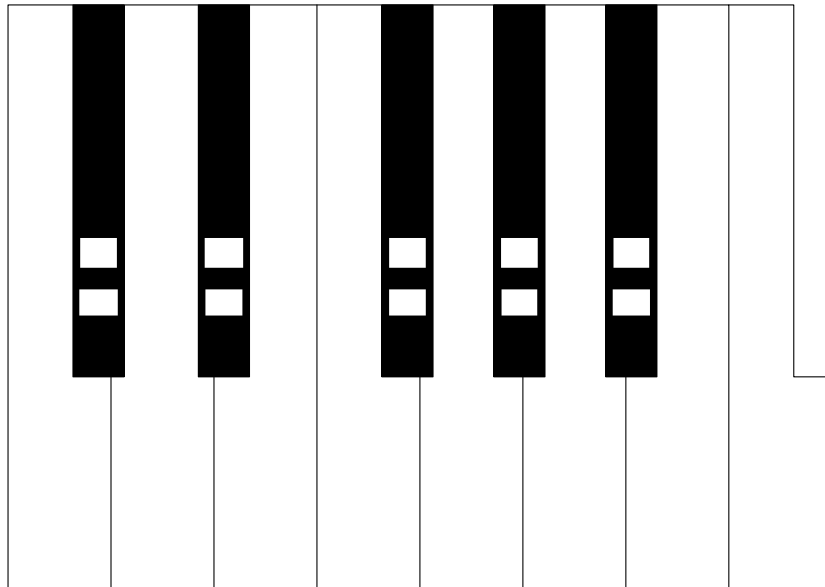
So the frequency is 50.05 Hz. To find the beat rate we take 0.05 and find its reciprocal which is 20. Now halve 20 to get a beat rate of $\frac{1}{2} \cdot 20 = 10$ Hz.

4.3 Homework

1. Label the white keys of the piano with the appropriate letters.



2. Label all of the keys of the piano with the appropriate letters.



3. For the following frequencies, find the frequencies of the seven octaves above it. Recall our notation for this is f_{oct} is the first octave above f , $f_{2.\text{oct}}$ is the second octave above f , etc.

- | | | |
|-----------------|------------------|------------------|
| (a) $f = 10$ Hz | (c) $f = 100$ Hz | (e) $f = 432$ Hz |
| (b) $f = 50$ Hz | (d) $f = 440$ Hz | (f) $f = 168$ Hz |

4. For the following frequencies, find the twelve fifths above it (find the fifth above the frequency f which we labeled as f_{fifth} , then the fifth above that frequency ($f_{2.\text{fifth}}$), etc. twelve times). Don't worry about keeping the frequencies within the one octave range.

- | | | |
|-----------------|------------------|------------------|
| (a) $f = 10$ Hz | (c) $f = 100$ Hz | (e) $f = 432$ Hz |
| (b) $f = 50$ Hz | (d) $f = 440$ Hz | (f) $f = 168$ Hz |

5. For the following frequencies, find the twelve fifths above it (find the fifth above the frequency f which we labeled as f_{fifth} , then the fifth above that frequency ($f_{2.\text{fifth}}$), etc twelve times). But this time keep the fifths within a one octave range of the fundamental frequency.

- | | | |
|-----------------|------------------|------------------|
| (a) $f = 10$ Hz | (c) $f = 100$ Hz | (e) $f = 432$ Hz |
| (b) $f = 50$ Hz | (d) $f = 440$ Hz | (f) $f = 168$ Hz |

6. Do the following:

- Find the twelfth fifth above $f = 27.5$ Hz, but don't rescale. Note that 27.5 Hz is an A (also called an A0).
- Looking at the circle of fifths, figure 4.3 on page 44, what note is this twelve fifth?
- Find the seventh octave above $f = 27.5$ Hz.
- What note is this seventh octave?
- What can we conclude from the four facts above?

7. Using the overtone series, show why the octave is the second most consonant note after the unison/tonic for the frequencies

- | | | |
|-----------------|------------------|------------------|
| (a) $f = 10$ Hz | (b) $f = 100$ Hz | (c) $f = 440$ Hz |
|-----------------|------------------|------------------|

8. Why does a note one octave above another sound the same except for the pitch?

- (a) $f = 10$ Hz
(b) $f = 100$ Hz
(c) $f = 440$ Hz

- Find the ratio for C , the fourth above it, F , and the fifth above it, G .
- Since we have to preserve octaves we can fill in the ratios for C , F , and G one octave higher.
- Next add the fifth above G which is a D and then the D one octave down.
- Now we find the perfect fifths above the D 's which are the A 's.
- Then the perfect fifth above the A which is the E and the E one octave down.
- Finally the perfect fifth above the E which is a B and the B one octave up.

[illegible]

11. Using the procedure below construct the ratios for the C Major scale using Just intonation.
- We start with the ratio for C and use the small integer ratios from the overtone series to get the fifth G , the fourth F , the third ($5 : 4$) E , and the sixth ($5 : 3$) A .
 - Go up an octave for each of these will give us the next line.
 - A fifth above an E is the B , so find that next and it's octave.
 - The final note is the D which we will make a fifth above G and then drop down an octave for the other D .

Step	C	D	E	F	G	A	B	C	D	E	F	G	A	B
1														
2														
3														
4														

12. What is the flaw in the Pythagorean and Just tuning systems? Now prove your answer.
13. In equal temperament, what value do we use for the fifth?
14. Using the Sum-to-product identity

$$\sin a + \sin b = 2 \sin \left(\frac{a+b}{2} \right) \cos \left(\frac{a-b}{2} \right)$$

find the frequency and beat rate for the following tones:

- $f_1 = 12$ Hz and $f_2 = 10$ Hz
- $f_1 = 440$ Hz and $f_2 = 438$ Hz
- $f_1 = 201$ Hz and $f_2 = 199$ Hz
- $f_1 = 100.5$ Hz and $f_2 = 100$ Hz
- $f_1 = 100.25$ Hz and $f_2 = 100$ Hz
- $f_1 = 100.125$ Hz and $f_2 = 100$ Hz

Appendix A

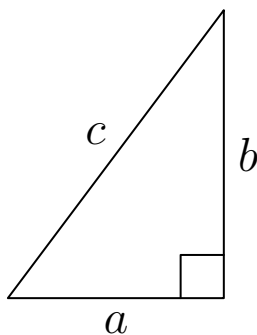
Answer key (partial)

A.1 Homework: Chapter 1

1. *Longitudinal waves.*
2. *Transverse waves.*

A.2 Homework: Chapter 2

1. Given the two sides of the right triangle below find the other side:

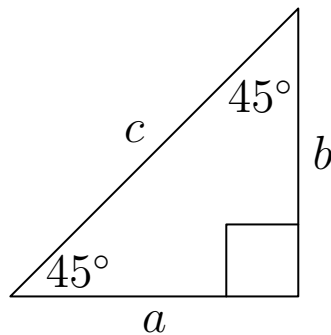


- (a) $c = 5$
- (b) $b = \sqrt{119}$
- (c) $a = 7$
- (d) $b = 15$
- (e)

- (f)
- (g)
- (h)
- (i)
- (j)

- (k)
- (l)
- (m) $a = \sqrt{99}$

2. Given the side below find the other two sides:



(a) $b = 3, c = 3\sqrt{2}$

(e) $a = 6, b = 6$

(i) $a = \frac{1}{\sqrt{2}}, b = \frac{1}{\sqrt{2}}$

(b)

(f)

(c) $a = 4, c = 4\sqrt{2}$

(g) $b = \frac{1}{2}, c = \frac{\sqrt{2}}{2}$

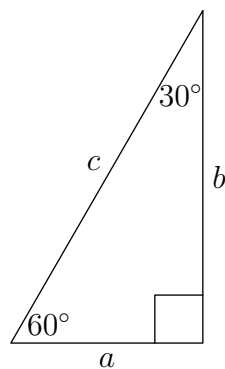
(j)

(d)

(h)

(k)

3. Given the side below find the other two sides:



(a) $b = 2\sqrt{3}, c = 4$

(e) $a = 3, b = 3\sqrt{3}$

(i) $a = \frac{1}{\sqrt{3}}, c = \frac{2}{\sqrt{3}}$

(b)

(f)

(j)

(c) $a = 4, c = 8$

(g)

(k) $a = \frac{1}{2\sqrt{3}}, c = \frac{1}{\sqrt{3}}$

(d)

(h)

4. (a) 60° (d) 30° (g) 60°
 (b) 45° (e) 45° (h) 45°
 (c) 30° (f) 60° (i) 30°

5. Find the following:

- | | | | |
|--------------------------|--------------------------|---------------------------|---------------------------|
| (a) 0 | (f) $\frac{\sqrt{3}}{2}$ | (j) $-\frac{1}{2}$ | (n) $-\frac{\sqrt{3}}{2}$ |
| (b) $\frac{1}{2}$ | (g) $\frac{1}{\sqrt{2}}$ | (k) $-\frac{1}{\sqrt{2}}$ | (o) $-\frac{1}{\sqrt{2}}$ |
| (c) $\frac{1}{\sqrt{2}}$ | (h) $\frac{1}{2}$ | (l) $-\frac{\sqrt{3}}{2}$ | (p) $-\frac{1}{2}$ |
| (d) $\frac{\sqrt{3}}{2}$ | (i) 0 | (m) -1 | (q) 0 |
| (e) 1 | | | |

6. Graph each of the following

- | | | |
|-----|-----|-----|
| (a) | (c) | (e) |
| (b) | (d) | (f) |

7. Find the following:

- | | | | |
|---------------------------|--------------------------|--------------------------|---------------------------|
| (a) $\frac{1}{2}$ | (c) $\frac{\sqrt{3}}{2}$ | (e) $\frac{\sqrt{3}}{2}$ | (h) $-\frac{1}{\sqrt{2}}$ |
| | | (f) 0 | |
| (b) $-\frac{1}{\sqrt{2}}$ | (d) $\frac{1}{2}$ | (g) -1 | (i) 0 |

8. Graph each of the following

- | | | |
|-----|-----|-----|
| (a) | (d) | (g) |
| (b) | (e) | |
| (c) | (f) | (h) |

A.3 Homework: Chapter 3

- The first overtone is the second harmonic, the second overtone is the third harmonic, etc.
- 100 Hz, 200 Hz, 300 Hz, 400 Hz, 500 Hz
 - 200 Hz, 400 Hz, 600 Hz, 800 Hz, 1000 Hz
 - 440 Hz, 880 Hz, 1320 Hz, 1760 Hz, 2200 Hz
 - 20 Hz, 40 Hz, 60 Hz, 80 Hz, 100 Hz

- (e) 50 Hz, 100 Hz, 150 Hz, 200 Hz, 250 Hz
- (f) 436 Hz, 872 Hz, 1308 Hz, 1744 Hz, 2180 Hz
- 3. (a) 10 Hz, 20 Hz, 30 Hz, 40 Hz, 50 Hz, ...
- (b) 15 Hz, 30 Hz, 45 Hz, 60 Hz, 75 Hz, ...
- (c)
- (d)
- (e)
- (f)
- (g) 1 Hz, 2 Hz, 3 Hz, 4 Hz, 5 Hz, ...
- (h) 27.5 Hz, 55 Hz, 82.5 Hz, 110 Hz, 137.5 Hz, ...
- (i)
- 4. (a) $F = 1000 \sin 440t + 300 \sin 880t + 100 \sin 1320$
- (b) $F = 100 \sin 440t + 400 \sin 880t + 200 \sin 1760$
- (c) $F = 700 \sin 100t + 1000 \sin 300t + 500 \sin 500$
- (d) $F = 1000 \sin 100t + 100 \sin 300t + 600 \sin 500t$
- (e) $F = 1000 \sin 100t + 700 \sin 200t + 300 \sin 300t + 500 \sin 400t + 100 \sin 500t$
- (f) $F = 300 \sin 100t + 1000 \sin 200t + 700 \sin 300t + 500 \sin 400t + 200 \sin 500t$
- 5. For the following Fourier series make the frequency response graphs.
- (a)
- (b)
- (c)
- (d)

A.4 Homework: Chapter 4

1. See figure 4.1.
2. See figure 4.2.
3. (a) $f_{2.\text{oct}} = 20 \text{ Hz}$, $f_{3.\text{oct}} = 40 \text{ Hz}$, $f_{4.\text{oct}} = 80 \text{ Hz}$, $f_{5.\text{oct}} = 160 \text{ Hz}$,
 $f_{6.\text{oct}} = 320 \text{ Hz}$, $f_{7.\text{oct}} = 640 \text{ Hz}$
- (b) $f_{2.\text{oct}} = 100 \text{ Hz}$, $f_{3.\text{oct}} = 200 \text{ Hz}$, $f_{4.\text{oct}} = 400 \text{ Hz}$, $f_{5.\text{oct}} = 800 \text{ Hz}$,
 $f_{6.\text{oct}} = 1600 \text{ Hz}$, $f_{7.\text{oct}} = 3200 \text{ Hz}$
- (c)
- (d) $f_{2.\text{oct}} = 880 \text{ Hz}$, $f_{3.\text{oct}} = 1760 \text{ Hz}$, $f_{4.\text{oct}} = 3520 \text{ Hz}$, $f_{5.\text{oct}} =$
 7040 Hz , $f_{6.\text{oct}} = 14080 \text{ Hz}$, $f_{7.\text{oct}} = 28160 \text{ Hz}$
- (e)

(f)

$$\begin{array}{lll}
f_{1.\text{fifth}} = 15 \text{ Hz}, & f_{2.\text{fifth}} = 22.5 \text{ Hz}, & f_{3.\text{fifth}} = 33.75 \text{ Hz}, \\
f_{4.\text{fifth}} = 50.63 \text{ Hz}, & f_{5.\text{fifth}} = 75.94 \text{ Hz}, & f_{6.\text{fifth}} = 113.91 \text{ Hz}, \\
f_{7.\text{fifth}} = 170.86 \text{ Hz}, & f_{8.\text{fifth}} = 256.29 \text{ Hz}, & f_{9.\text{fifth}} = 384.43 \text{ Hz}, \\
f_{10.\text{fifth}} = 576.65 \text{ Hz}, & f_{11.\text{fifth}} = 864.98 \text{ Hz}, & f_{12.\text{fifth}} = 1297.46 \text{ Hz}
\end{array}$$

(b)

$$\begin{array}{lll}
f_{1.\text{fifth}} = 105 \text{ Hz}, & f_{2.\text{fifth}} = 225 \text{ Hz}, & f_{3.\text{fifth}} = 337.5 \text{ Hz}, \\
f_{4.\text{fifth}} = 506.25 \text{ Hz}, & f_{5.\text{fifth}} = 759.38 \text{ Hz}, & f_{6.\text{fifth}} = 1139.06 \text{ Hz}, \\
f_{7.\text{fifth}} = 1708.59 \text{ Hz}, & f_{8.\text{fifth}} = 2562.89 \text{ Hz}, & f_{9.\text{fifth}} = 3844.34 \text{ Hz}, \\
f_{10.\text{fifth}} = 5766.50 \text{ Hz}, & f_{11.\text{fifth}} = 8649.76 \text{ Hz}, & f_{12.\text{fifth}} = 12974.63 \text{ Hz}
\end{array}$$

$$\begin{array}{lll}
f_{1.\text{fifth}} = 660 \text{ Hz}, & f_{2.\text{fifth}} = 990 \text{ Hz}, & f_{3.\text{fifth}} = 1485 \text{ Hz}, \\
f_{4.\text{fifth}} = 2227.5 \text{ Hz}, & f_{5.\text{fifth}} = 3341.25 \text{ Hz}, & f_{6.\text{fifth}} = 5011.88 \text{ Hz}, \\
f_{7.\text{fifth}} = 7517.81 \text{ Hz}, & f_{8.\text{fifth}} = 11276.72 \text{ Hz}, & f_{9.\text{fifth}} = 16915.08 \text{ Hz}, \\
f_{10.\text{fifth}} = 25372.62 \text{ Hz}, & f_{11.\text{fifth}} = 38058.93 \text{ Hz}, & f_{12.\text{fifth}} = 57088.39 \text{ Hz}
\end{array}$$

(e)

(f)

$$\begin{array}{lll}
f_{1.\text{fifth}} = 15 \text{ Hz}, & f_{2.\text{fifth}} = 11.25 \text{ Hz}, & f_{3.\text{fifth}} = 16.88 \text{ Hz}, \\
f_{4.\text{fifth}} = 12.66 \text{ Hz}, & f_{5.\text{fifth}} = 18.98 \text{ Hz}, & f_{6.\text{fifth}} = 14.24 \text{ Hz}, \\
f_{7.\text{fifth}} = 10.68 \text{ Hz}, & f_{8.\text{fifth}} = 16.02 \text{ Hz}, & f_{9.\text{fifth}} = 12.01 \text{ Hz}, \\
f_{10.\text{fifth}} = 18.02 \text{ Hz}, & f_{11.\text{fifth}} = 13.52 \text{ Hz}, & f_{12.\text{fifth}} = 10.14 \text{ Hz}
\end{array}$$

(b)

(c)

(d)

(e)

(f)

6. Do the following:

(a) 3568.02 Hz

(b) A.

(c) 3520 Hz

(d) A.

(e) Going up twelve fifths is not the same as going up seven octaves.

	Unison	Octave
	10 Hz	
	20 Hz	20 Hz
	30 Hz	
7. (a)	40 Hz	40 Hz
	50 Hz	
	60 Hz	60 Hz
	70 Hz	
	80 Hz	80 Hz

	Unison	Octave		Unison	Octave
	100 Hz			440 Hz	
	200 Hz	200 Hz		880 Hz	880 Hz
	300 Hz			1320 Hz	
(b)	400 Hz	400 Hz	(c)	1760 Hz	1760 Hz
	500 Hz			2200 Hz	
	600 Hz	600 Hz		2640 Hz	2640 Hz
	700 Hz			3080 Hz	
	800 Hz	800 Hz		35200 Hz	35200 Hz

8. The pitch one octave up contains every other frequency from the fundamental frequencies overtone series.

	Unison	Fifth		Unison	Fifth		Unison	Fifth
	10 Hz			100 Hz			440 Hz	
		15 Hz			150 Hz			660 Hz
	20 Hz			200 Hz			880 Hz	
9. (a)	30 Hz	30 Hz	(b)	300 Hz	300 Hz	(c)	1320 Hz	1320 Hz
	40 Hz			400 Hz			1760 Hz	
		45 Hz			450 Hz			1980 Hz
	50 Hz			500 Hz			2200 Hz	
	60 Hz	60 Hz		600 Hz	600 Hz		2640 Hz	2640 Hz

10. See table 4.2.

11. See table 4.5.

12. The octaves and fifths do not agree in terms of frequencies.

The proof is $\left(\frac{3}{2}\right)^{12} = 129.75 \neq 128 = 2^7$.

13. The fifth is defined to be $r = \sqrt[12]{2^7} = 2^{\frac{7}{12}}$.

14. (a) $f = 11$, $\text{br} = \frac{1}{2}$
 (b) $f = 399$, $\text{br} = \frac{1}{2}$
 (c) $f = 200$, $\text{br} = \frac{1}{2}$
 (d) $f = 100.25$, $\text{br} = 2$
 (e) $f = 100.125$, $\text{br} = 4$
 (f) $f = 100.0625$, $\text{br} = 8$

Appendix B

Addition figures



Figure B.1: Piano key diagram.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
<i>G</i>	<i>G</i> $\sharp$	<i>A</i>	<i>A</i> $\sharp$	<i>B</i>	<i>C</i>	<i>C</i> $\sharp$	<i>D</i>	<i>D</i> $\sharp$	<i>E</i>	<i>F</i>	<i>F</i> $\sharp$	<i>G</i>	<i>G</i> $\sharp$	<i>A</i>	<i>A</i> $\sharp$	<i>B</i>	<i>C</i>	<i>C</i> $\sharp$	<i>D</i>	<i>D</i> $\sharp$
<i>D</i>	<i>D</i> $\sharp$	<i>E</i>	<i>F</i>	<i>F</i> $\sharp$	<i>G</i>	<i>G</i> $\sharp$	<i>A</i>	<i>A</i> $\sharp$	<i>B</i>	<i>C</i>	<i>C</i> $\sharp$	<i>D</i>	<i>D</i> $\sharp$	<i>E</i>	<i>F</i>	<i>F</i> $\sharp$	<i>G</i>	<i>G</i> $\sharp$	<i>A</i>	<i>A</i> $\sharp$
<i>A</i>	<i>A</i> $\sharp$	<i>B</i>	<i>C</i>	<i>C</i> $\sharp$	<i>D</i>	<i>D</i> $\sharp$	<i>E</i>	<i>F</i>	<i>F</i> $\sharp$	<i>G</i>	<i>G</i> $\sharp$	<i>A</i>	<i>A</i> $\sharp$	<i>B</i>	<i>C</i>	<i>C</i> $\sharp$	<i>D</i>	<i>D</i> $\sharp$	<i>E</i>	<i>F</i>
<i>E</i>	<i>F</i>	<i>F</i> $\sharp$	<i>G</i>	<i>G</i> $\sharp$	<i>A</i>	<i>A</i> $\sharp$	<i>B</i>	<i>C</i>	<i>C</i> $\sharp$	<i>D</i>	<i>D</i> $\sharp$	<i>E</i>	<i>F</i>	<i>F</i> $\sharp$	<i>G</i>	<i>G</i> $\sharp$	<i>A</i>	<i>A</i> $\sharp$	<i>B</i>	<i>C</i>
			●		●		●		●			●●				●		●		

Figure B.2: Bass guitar fret board with tones.

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